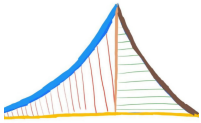
Math for EE



G. V. V. Sharma

About this Book

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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First manual appeared in January 2018
First edition published on July 10, 2024

In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. 10

A. Formulae

1.1.1 A tangent to a circle with centre $\mathbf{O}$ is perpendicular to the radius through the point of contact $\mathbf{T}$,

$$(\mathbf{P}-\mathbf{T})^{\top}(\mathbf{T}-\mathbf{O})=0 \tag{1.1.1.1}$$

where $\mathbf{P}$ is any other point on the tangent.

1.1.2 The lengths of the two tangents drawn from an external point $\mathbf{P}$ to a circle of radius a and centre $\mathbf{O}$ are equal and

$$\|\mathbf{P}-\mathbf{T}_1\|=\|\mathbf{P}-\mathbf{T}_2\|=\sqrt{\|\mathbf{P}-\mathbf{O}\|^2-a^2} \tag{1.1.2.1}$$

where $\mathbf{T}_1$ and $\mathbf{T}_2$ are the points of contact.

Solution: From (1.1.1.1) and Pythagoras' theorem in the right triangle $\mathbf{OT}_i\mathbf{P}$, $i=1,2$,

$$\|\mathbf{P}-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+\|\mathbf{T}_i-\mathbf{O}\|^2=\|\mathbf{P}-\mathbf{T}_i\|^2+a^2 \tag{1.1.2.2}$$

which is the same for $i=1,2$, yielding (1.1.2.1).

1.1.3 In a right triangle with an acute angle θ,

$$\sin\theta=\frac{\text{opposite}}{\text{hypotenuse}},\quad \tan\theta=\frac{\text{opposite}}{\text{adjacent}} \tag{1.1.3.1}$$

1.1.4 The angle θ between the vectors $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\cos\theta=\frac{\mathbf{a}^{\top}\mathbf{b}}{\|\mathbf{a}\|\,\|\mathbf{b}\|} \tag{1.1.4.1}$$

For complementary angles,

$$\cos(90^\circ-\theta)=\sin\theta,\quad \cos^2\theta+\cos^2(90^\circ-\theta)=1 \tag{1.1.4.2}$$

1.1.5 The locus of a point $\mathbf{x}$ equidistant from the points $\mathbf{A}$ and $\mathbf{B}$ is

$$(\mathbf{A}-\mathbf{B})^{\top}\mathbf{x}=\frac{\|\mathbf{A}\|^2-\|\mathbf{B}\|^2}{2} \tag{1.1.5.1}$$

Solution:

$$\|\mathbf{x}-\mathbf{A}\|^2=\|\mathbf{x}-\mathbf{B}\|^2 \tag{1.1.5.2}$$

$$\implies \|\mathbf{x}\|^2-2\mathbf{A}^{\top}\mathbf{x}+\|\mathbf{A}\|^2=\|\mathbf{x}\|^2-2\mathbf{B}^{\top}\mathbf{x}+\|\mathbf{B}\|^2 \tag{1.1.5.3}$$

which simplifies to (1.1.5.1).

1.1.6 For the quadratic equation $px^2+qx+r=0$ with roots α and β,

$$\alpha+\beta=-\frac{q}{p},\quad \alpha\beta=\frac{r}{p} \tag{1.1.6.1}$$

1.1.7 The nth term of an A.P. with first term a and common difference d is

$$a_n=a+(n-1)\,d \tag{1.1.7.1}$$

1.1.8 If $\mathbf{R}$ lies on the line through $\mathbf{P}$ and $\mathbf{Q}$, then

$$\mathbf{R}-\mathbf{P}=\lambda\left(\mathbf{Q}-\mathbf{P}\right) \tag{1.1.8.1}$$

for some scalar λ, and $\mathbf{R}$ divides the line segment $\mathbf{PQ}$ in the ratio

$$\mathbf{PR}:\mathbf{RQ}=\lambda:(1-\lambda) \tag{1.1.8.2}$$

Solution: From (1.1.9.1), the columns of $\begin{pmatrix}\mathbf{Q}-\mathbf{P} & \mathbf{R}-\mathbf{P}\end{pmatrix}$ are linearly dependent, which gives the first equation. Then

$$\mathbf{R}-\mathbf{Q}=(\lambda-1)\left(\mathbf{Q}-\mathbf{P}\right) \tag{1.1.8.3}$$

so $\|\mathbf{R}-\mathbf{P}\|:\|\mathbf{R}-\mathbf{Q}\|=|\lambda|:|1-\lambda|$, yielding the ratio.

1.1.9 The points $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are collinear if and only if

$$\operatorname{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}=1 \tag{1.1.9.1}$$

1.1.10 The roots of the quadratic equation $px^2+qx+r=0$ are

$$x=\frac{-q\pm\sqrt{q^2-4pr}}{2p} \tag{1.1.10.1}$$

The roots are equal if and only if

$$q^2-4pr=0 \tag{1.1.10.2}$$

1.1.11 The equation of a conic is

$$\mathbf{x}^{\top}\mathbf{V}\mathbf{x}+2\mathbf{u}^{\top}\mathbf{x}+f=0 \tag{1.1.11.1}$$

where $\mathbf{V}$ is a symmetric matrix. It is a parabola if $\operatorname{rank}(\mathbf{V})=1$. The points of intersection with the line $\mathbf{x}=\mathbf{h}+\kappa\mathbf{m}$ satisfy

$$\kappa^2\mathbf{m}^{\top}\mathbf{V}\mathbf{m}+2\kappa\mathbf{m}^{\top}\left(\mathbf{V}\mathbf{h}+\mathbf{u}\right)+g\left(\mathbf{h}\right)=0 \tag{1.1.11.2}$$

1.1.12 The equation of a circle with centre $\mathbf{O}$ and radius r is

$$\|\mathbf{x}\|^2+2\mathbf{u}^{\top}\mathbf{x}+f=0 \tag{1.1.12.1}$$

where

$$\mathbf{u}=-\mathbf{O},\quad f=\|\mathbf{O}\|^2-r^2 \tag{1.1.12.2}$$

This is (1.1.11.1) with $\mathbf{V}=\mathbf{I}$. The point $\mathbf{h}$ is external to the circle if

$$g\left(\mathbf{h}\right)=\|\mathbf{h}\|^2+2\mathbf{u}^{\top}\mathbf{h}+f>0 \tag{1.1.12.3}$$

1.1.13 For an external point $\mathbf{h}$ of the circle in (1.1.12.1), define

$$\mathbf{\Sigma}=\left(\mathbf{h}+\mathbf{u}\right)\left(\mathbf{h}+\mathbf{u}\right)^{\top}-g\left(\mathbf{h}\right)\mathbf{I} \tag{1.1.13.1}$$

Let λ_1, λ_2 be the roots of $\left|\lambda\mathbf{I}-\mathbf{\Sigma}\right|=0$ (the eigenvalues) and $\mathbf{p}_1$, $\mathbf{p}_2$ the

corresponding eigenvectors, found by row reduction of $\mathbf{\Sigma p} = \lambda \mathbf{p}$. Define

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix} \frac{\mathbf{p}_1}{\|\mathbf{p}_1\|} & \frac{\mathbf{p}_2}{\|\mathbf{p}_2\|} \end{pmatrix} \tag{1.1.13.2}$$

Then $\mathbf{P}^\top = \mathbf{P}^{-1}$ and $\mathbf{P}^\top \mathbf{\Sigma P} = \mathbf{D}$. The direction vectors of the tangents from $\mathbf{h}$ are

$$\mathbf{m} = \mathbf{P} \begin{pmatrix} \sqrt{|\lambda_2|} \\ \pm\sqrt{|\lambda_1|} \end{pmatrix} \tag{1.1.13.3}$$

and the points of contact are

$$\mathbf{x} = \mathbf{h} + \mu \mathbf{m}, \quad \mu = -\frac{\mathbf{m}^\top (\mathbf{h} + \mathbf{u})}{\mathbf{m}^\top \mathbf{m}} \tag{1.1.13.4}$$

Solution: The line $\mathbf{x} = \mathbf{h} + \mu \mathbf{m}$ is tangent to the circle if (1.1.11.2), with $\mathbf{V} = \mathbf{I}$, has equal roots,

$$\left(\mathbf{m}^\top (\mathbf{h} + \mathbf{u})\right)^2 - g(\mathbf{h}) \mathbf{m}^\top \mathbf{m} = \mathbf{m}^\top \mathbf{\Sigma m} = 0 \tag{1.1.13.5}$$

Substituting $\mathbf{m} = \mathbf{P y}$ and using $\mathbf{P}^\top \mathbf{\Sigma P} = \mathbf{D}$,

$$\lambda_1 y_1^2 + \lambda_2 y_2^2 = 0 \tag{1.1.13.6}$$

Since λ_1 and λ_2 have opposite signs, this is satisfied by $\mathbf{y} = \begin{pmatrix} \sqrt{|\lambda_2|} \\ \pm\sqrt{|\lambda_1|} \end{pmatrix}$, yielding (1.1.13.3). The equal root of (1.1.11.2) is $\mu = -\frac{\mathbf{m}^\top (\mathbf{h}+\mathbf{u})}{\mathbf{m}^\top \mathbf{m}}$, yielding (1.1.13.4). where $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V h} + 2\mathbf{u}^\top \mathbf{h} + f$. **Solution:** Substituting $\mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ in (1.1.11.1) and using $\mathbf{V}^\top = \mathbf{V}$,

$$(\mathbf{h} + \kappa \mathbf{m})^\top \mathbf{V} (\mathbf{h} + \kappa \mathbf{m}) + 2\mathbf{u}^\top (\mathbf{h} + \kappa \mathbf{m}) + f = 0 \tag{1.1.13.7}$$

which on expanding yields (1.1.11.2).

1.1.14 The unit step function is

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \tag{1.1.14.1}$$

An A.P. with first term a and common difference d, with the index starting from 0, is

$$x(n) = (a + nd)\, u(n) \tag{1.1.14.2}$$

where $x(n)$ is the $(n+1)^{\text{th}}$ term.

1.1.15 The Z-transform of a sequence $x(n)$ is

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n} \tag{1.1.15.1}$$

For convolution, delay and accumulation,

$$x_1(n) * x_2(n) \overset{\mathcal{Z}}{\longleftrightarrow} X_1(z) X_2(z), \quad x(n-1) \overset{\mathcal{Z}}{\longleftrightarrow} z^{-1} X(z) \tag{1.1.15.2}$$

The unit step $u(n)$ satisfies

$$u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{1-z^{-1}}, \quad |z| > 1 \tag{1.1.15.3}$$

Since $\sum_{k=0}^{n} x(k) = x(n) * u(n)$, the running sum satisfies

$$\sum_{k=0}^{n} x(k) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{X(z)}{1-z^{-1}} \tag{1.1.15.4}$$

1.1.16 For $|z| > 1$,

$$n\,u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{z^{-1}}{\left(1-z^{-1}\right)^2}, \quad \frac{(n+1)(n+2)}{2} u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1-z^{-1}\right)^3} \tag{1.1.16.1}$$

Solution: Since $\sum_{k=0}^{n} 1 = n+1$, from (1.1.15.2),

$$(n+1)\,u(n) = u(n) * u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1-z^{-1}\right)^2} \tag{1.1.16.2}$$

Since $\sum_{k=0}^{n} (k+1) = \frac{(n+1)(n+2)}{2}$,

$$\frac{(n+1)(n+2)}{2} u(n) = (n+1)\,u(n) * u(n) \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1}{\left(1-z^{-1}\right)^3} \tag{1.1.16.3}$$

Delaying $(n+1)\,u(n)$ by one sample gives $n\,u(n-1) = n\,u(n)$, which has the transform $\frac{z^{-1}}{(1-z^{-1})^2}$ from (1.1.15.2).

1.1.17 For an event E and its complement E',

$$\Pr(E) + \Pr(E') = 1 \tag{1.1.17.1}$$

1.1.18 Let $X_1, X_2 \in \{1,2,3,4,5,6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF)

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{1.1.18.1}$$

Let $X = X_1 + X_2$ denote their sum. The PMF of X is the discrete convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$,

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k p_{X_1}(n-k)\, p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \tag{1.1.18.2}$$

Substituting (1.1.18.1) into (1.1.18.2) yields the triangular PMF

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \le n \le 7 \\ \frac{13-n}{36}, & 7 < n \le 12 \\ 0, & n > 12 \end{cases} \tag{1.1.18.3}$$

The following code verifies (1.1.18.3) in Fig. 1.1.18.1

codes/dice_pmf.py

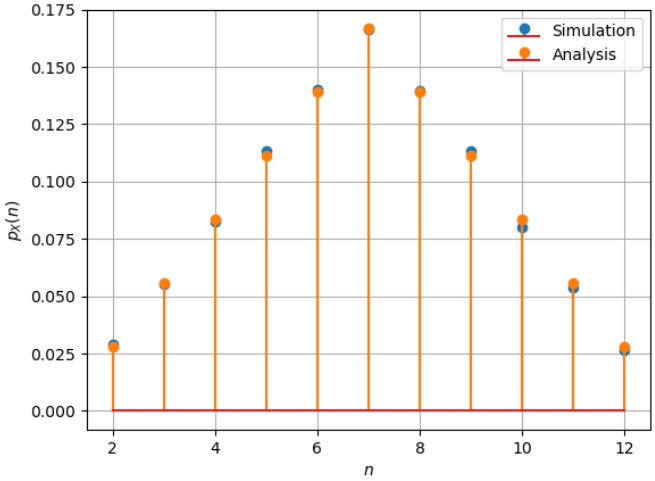


Fig. 1.1.18.1: PMF of the sum of two fair dice: Simulation vs Analysis

1.1.19 If a person completes a work in t days, the rate of work is $\frac{1}{t}$ of the work per day. The rates of persons working together add up,

$$\frac{1}{t} = \frac{1}{t_1} + \frac{1}{t_2} \tag{1.1.19.1}$$

1.1.20 The angle of depression of an object $\mathbf{C}$ from a point $\mathbf{T}$ is equal to the angle of elevation of $\mathbf{T}$ from $\mathbf{C}$ (alternate angles). If $\mathbf{O}$ is the foot of the perpendicular from $\mathbf{T}$ to the ground, this angle θ is the angle between $\mathbf{T}-\mathbf{C}$ and $\mathbf{O}-\mathbf{C}$. From (1.1.4.1),

$$\cos\theta = \frac{(\mathbf{T}-\mathbf{C})^{\top}(\mathbf{O}-\mathbf{C})}{\|\mathbf{T}-\mathbf{C}\|\,\|\mathbf{O}-\mathbf{C}\|} \tag{1.1.20.1}$$

1.1.21 The points on the internal bisector of the angle at a vertex $\mathbf{A}$, with adjacent vertices

$\mathbf{B}$ and $\mathbf{D}$, are

$$\mathbf{x} = \mathbf{A} + t\left(\frac{\mathbf{B}-\mathbf{A}}{\|\mathbf{B}-\mathbf{A}\|} + \frac{\mathbf{D}-\mathbf{A}}{\|\mathbf{D}-\mathbf{A}\|}\right), \quad t \in \mathbb{R} \tag{1.1.21.1}$$

Solution: The two terms in the bracket are unit vectors along the sides $\mathbf{AB}$ and $\mathbf{AD}$. Their sum is the diagonal of a rhombus, which bisects the angle between them, yielding (1.1.21.1).

1.1.22 The distance of a point $\mathbf{O}$ from the line passing through $\mathbf{P}$ with normal vector $\mathbf{n}$ is

$$d = \frac{\left|\mathbf{n}^\top(\mathbf{O}-\mathbf{P})\right|}{\|\mathbf{n}\|} \tag{1.1.22.1}$$

Solution: The distance is the length of the projection of $\mathbf{O}-\mathbf{P}$ on $\mathbf{n}$, yielding (1.1.22.1).

B. Problems

1.2.1 If the angle between two tangents drawn from an external point P to a circle of radius a and centre O, is $60°$, then find the length of OP. (10, 2017)

Solution: Let $\mathbf{T}_1$ and $\mathbf{T}_2$ be the points of contact of the tangents from $\mathbf{P}$. From (1.1.2.1),

$$\|\mathbf{P}-\mathbf{T}_1\| = \|\mathbf{P}-\mathbf{T}_2\| \tag{1.2.1.1}$$

Also $\|\mathbf{O}-\mathbf{T}_1\| = \|\mathbf{O}-\mathbf{T}_2\| = a$ and $\mathbf{OP}$ is common. So the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are congruent, and $\mathbf{OP}$ bisects the angle between the tangents,

$$\angle\mathbf{OPT}_1 = \frac{60°}{2} = 30° \tag{1.2.1.2}$$

From (1.1.1.1), the triangle $\mathbf{OT}_1\mathbf{P}$ is right angled at $\mathbf{T}_1$. Using (1.1.3.1),

$$\sin 30° = \frac{a}{\|\mathbf{O}-\mathbf{P}\|} \tag{1.2.1.3}$$

$$\implies \|\mathbf{O}-\mathbf{P}\| = \frac{a}{\frac{1}{2}} = 2a \tag{1.2.1.4}$$

To verify numerically, take $a = 3$ and

$$\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix}6\\0\end{pmatrix} \tag{1.2.1.5}$$

so that $\|\mathbf{O}-\mathbf{P}\| = 6 = 2a$. Let $\mathbf{T}$ be a point of contact. From (1.1.1.1),

$$(\mathbf{P}-\mathbf{T})^\top\mathbf{T} = 0 \implies \mathbf{P}^\top\mathbf{T} = \|\mathbf{T}\|^2 = a^2 = 9 \tag{1.2.1.6}$$

With $\mathbf{T} = \begin{pmatrix}x\\y\end{pmatrix}$,

$$6x = 9 \implies x = \frac{3}{2}, \quad y^2 = 9 - \frac{9}{4} = \frac{27}{4} \implies y = \pm\frac{3\sqrt{3}}{2} \tag{1.2.1.7}$$

Hence, the points of contact are

$$\mathbf{T}_1 = \begin{pmatrix} \frac{3}{2} \\ \frac{3\sqrt{3}}{2} \end{pmatrix}, \quad \mathbf{T}_2 = \begin{pmatrix} \frac{3}{2} \\ -\frac{3\sqrt{3}}{2} \end{pmatrix} \tag{1.2.1.8}$$

The lengths of the tangents are

$$\|\mathbf{P} - \mathbf{T}_1\| = \left\| \begin{pmatrix} \frac{9}{2} \\ -\frac{3\sqrt{3}}{2} \end{pmatrix} \right\| = \sqrt{\frac{81}{4} + \frac{27}{4}} = 3\sqrt{3} = \|\mathbf{P} - \mathbf{T}_2\| \tag{1.2.1.9}$$

which are equal and agree with $\sqrt{\|\mathbf{P} - \mathbf{O}\|^2 - a^2} = \sqrt{36 - 9} = 3\sqrt{3}$ from (1.1.2.1). The angle between the tangents is the angle between $\mathbf{T}_1 - \mathbf{P}$ and $\mathbf{T}_2 - \mathbf{P}$. From (1.1.4.1),

$$\cos\theta = \frac{(\mathbf{T}_1 - \mathbf{P})^\top (\mathbf{T}_2 - \mathbf{P})}{\|\mathbf{T}_1 - \mathbf{P}\| \, \|\mathbf{T}_2 - \mathbf{P}\|} = \frac{\frac{81}{4} - \frac{27}{4}}{27} = \frac{1}{2} \tag{1.2.1.10}$$

Hence,

$$\theta = 60^\circ \tag{1.2.1.11}$$

which verifies that $OP = 2a = 6$ for $a = 3$. The following code generates Fig. 1.2.1.1

codes/tangent_length.py

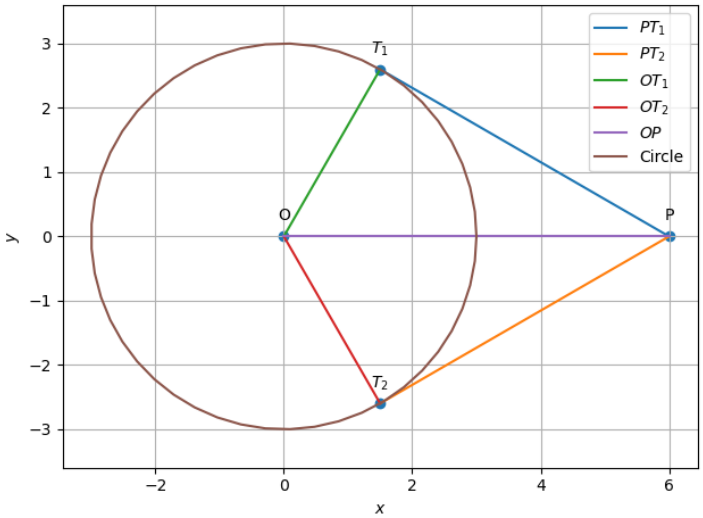


Fig. 1.2.1.1: Tangents from $\mathbf{P}$ to the circle of radius $a = 3$

1.2.2 A circle touches all the four sides of a quadrilateral ABCD. Prove that

$$AB + CD = BC + DA$$

(10, 2017)

Solution: Let the circle touch AB, BC, CD and DA at $\mathbf{P}$, $\mathbf{Q}$, $\mathbf{R}$ and $\mathbf{S}$ respectively. The tangents drawn from each of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ are equal, from (1.1.2.1). Hence,

$$AP = AS, \quad BP = BQ, \quad CR = CQ, \quad DR = DS \tag{1.2.2.1}$$

Adding the above,

$$(AP + BP) + (CR + DR) = (AS + DS) + (BQ + CQ) \tag{1.2.2.2}$$

Since $\mathbf{P}$ lies on AB, $\mathbf{R}$ on CD, $\mathbf{S}$ on DA and $\mathbf{Q}$ on BC,

$$AP + BP = AB, \quad CR + DR = CD, \quad AS + DS = DA, \quad BQ + CQ = BC \tag{1.2.2.3}$$

Substituting in the above,

$$AB + CD = BC + DA \tag{1.2.2.4}$$

To verify numerically, choose

$$\mathbf{A} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 6 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} \tag{1.2.2.5}$$

The lengths of the sides are

$$AB = \|\mathbf{B} - \mathbf{A}\| = 6, \quad BC = \|\mathbf{C} - \mathbf{B}\| = 5, \quad CD = \|\mathbf{D} - \mathbf{C}\| = 3, \quad DA = \|\mathbf{A} - \mathbf{D}\| \tag{1.2.2.6}$$

so that

$$AB + CD = 9 = BC + DA \tag{1.2.2.7}$$

and the given condition is satisfied. The centre $\mathbf{O}$ of the circle lies on the bisectors of the angles at $\mathbf{A}$ and $\mathbf{B}$. From (1.1.21.1), the directions of these bisectors are

$$\mathbf{m}_A = \frac{\mathbf{B} - \mathbf{A}}{\|\mathbf{B} - \mathbf{A}\|} + \frac{\mathbf{D} - \mathbf{A}}{\|\mathbf{D} - \mathbf{A}\|} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} \tag{1.2.2.8}$$

$$\mathbf{m}_B = \frac{\mathbf{A} - \mathbf{B}}{\|\mathbf{A} - \mathbf{B}\|} + \frac{\mathbf{C} - \mathbf{B}}{\|\mathbf{C} - \mathbf{B}\|} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} + \begin{pmatrix} -\frac{3}{5} \\ \frac{4}{5} \end{pmatrix} = \frac{4}{5}\begin{pmatrix} -2 \\ 1 \end{pmatrix} \tag{1.2.2.9}$$

Let $\mathbf{O} = \mathbf{A} + t\mathbf{m}_A = \mathbf{B} + s\begin{pmatrix} -2 \\ 1 \end{pmatrix}$. Then

$$t\begin{pmatrix} 1 \\ 1 \end{pmatrix} + s\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \mathbf{B} - \mathbf{A} = \begin{pmatrix} 6 \\ 0 \end{pmatrix} \tag{1.2.2.10}$$

The augmented matrix is

$$\begin{pmatrix} 1 & 2 & 6 \\ 1 & -1 & 0 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2 - R_1} \begin{pmatrix} 1 & 2 & 6 \\ 0 & -3 & -6 \end{pmatrix} \tag{1.2.2.11}$$

which gives $s = 2$ and $t = 2$. Therefore,

$$\mathbf{O} = \mathbf{A} + 2\begin{pmatrix}1\\1\end{pmatrix} = \begin{pmatrix}2\\2\end{pmatrix} \tag{1.2.2.12}$$

The radius is the distance of $\mathbf{O}$ from the side AB, which has the normal vector $\mathbf{n}_1 = \begin{pmatrix}0\\1\end{pmatrix}$. From (1.1.22.1),

$$r = \frac{\left|\mathbf{n}_1^\top(\mathbf{O}-\mathbf{A})\right|}{\|\mathbf{n}_1\|} = \frac{\left|\begin{pmatrix}0 & 1\end{pmatrix}\begin{pmatrix}2\\2\end{pmatrix}\right|}{1} = 2 \tag{1.2.2.13}$$

To check that the circle touches the other three sides, the normal vectors of BC, CD and DA are $\mathbf{n}_2 = \begin{pmatrix}4\\3\end{pmatrix}$ (since $\mathbf{n}_2^\top(\mathbf{C}-\mathbf{B}) = 0$), $\mathbf{n}_3 = \begin{pmatrix}0\\1\end{pmatrix}$ and $\mathbf{n}_4 = \begin{pmatrix}1\\0\end{pmatrix}$. Then

$$\frac{\left|\mathbf{n}_2^\top(\mathbf{O}-\mathbf{B})\right|}{\|\mathbf{n}_2\|} = \frac{\left|\begin{pmatrix}4 & 3\end{pmatrix}\begin{pmatrix}-4\\2\end{pmatrix}\right|}{5} = 2 \tag{1.2.2.14}$$

$$\frac{\left|\mathbf{n}_3^\top(\mathbf{O}-\mathbf{C})\right|}{\|\mathbf{n}_3\|} = \left|\begin{pmatrix}0 & 1\end{pmatrix}\begin{pmatrix}-1\\-2\end{pmatrix}\right| = 2 \tag{1.2.2.15}$$

$$\frac{\left|\mathbf{n}_4^\top(\mathbf{O}-\mathbf{D})\right|}{\|\mathbf{n}_4\|} = \left|\begin{pmatrix}1 & 0\end{pmatrix}\begin{pmatrix}2\\-2\end{pmatrix}\right| = 2 \tag{1.2.2.16}$$

Hence, the circle with centre $\mathbf{O} = \begin{pmatrix}2\\2\end{pmatrix}$ and radius $r = 2$ touches all four sides of the quadrilateral $ABCD$. The following code generates Fig. 1.2.2.1

codes/quad_incircle.py

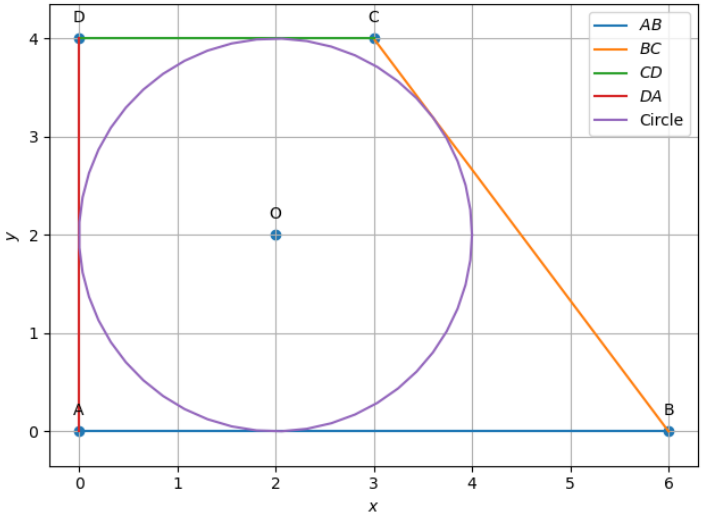


Fig. 1.2.2.1: Circle touching all four sides of the quadrilateral $ABCD$

1.2.3 If the distances of $\mathbf{P}\,(x, y)$ from $\mathbf{A}\,(5, 1)$ and $\mathbf{B}\,(-1, 5)$ are equal, then prove that $3x = 2y$. (10, 2017)

Solution: Let

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{A} = \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} -1 \\ 5 \end{pmatrix} \tag{1.2.3.1}$$

Then

$$\mathbf{A} - \mathbf{B} = \begin{pmatrix} 6 \\ -4 \end{pmatrix}, \quad \|\mathbf{A}\|^2 = 26, \quad \|\mathbf{B}\|^2 = 26 \tag{1.2.3.2}$$

Substituting in (1.1.5.1),

$$\begin{pmatrix} 6 & -4 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \frac{26 - 26}{2} = 0 \tag{1.2.3.3}$$

$$\implies 6x - 4y = 0 \tag{1.2.3.4}$$

$$\therefore 3x = 2y \tag{1.2.3.5}$$

The following code generates Fig. 1.2.3.1

codes/perp_bisector.py

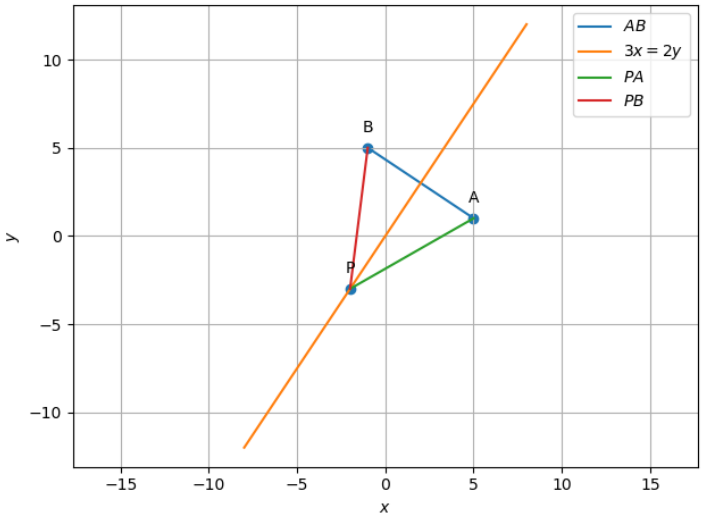


Fig. 1.2.3.1: Perpendicular bisector of AB

1.2.4 Find the value of p, for which one root of the quadratic equation $px^2 - 14x + 8 = 0$ is 6 times the other. (10, 2017)

Solution: Let the roots be α and 6α. From (1.1.6.1),

$$\alpha + 6\alpha = \frac{14}{p} \tag{1.2.4.1}$$

$$\alpha\,(6\alpha) = \frac{8}{p} \tag{1.2.4.2}$$

From (1.2.4.1),

$$\alpha = \frac{2}{p} \tag{1.2.4.3}$$

Substituting in (1.2.4.2),

$$\frac{24}{p^2} = \frac{8}{p} \tag{1.2.4.4}$$

$$\implies p = 3 \tag{1.2.4.5}$$

For $p = 3$, the roots are 4 and $\frac{2}{3}$, and indeed $4 = 6\left(\frac{2}{3}\right)$. For $p = 3$, consider the parabola $y = 3x^2 - 14x + 8$. Its quadratic form is

$$\mathbf{x}^\top \mathbf{V}\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0 \tag{1.2.4.6}$$

where

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} -7 \\ -\frac{1}{2} \end{pmatrix}, \quad f = 8 \tag{1.2.4.7}$$

Since rank $(\mathbf{V}) = 1$, this is a parabola. The x-axis is the line $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ with

$$\mathbf{h} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{1.2.4.8}$$

Then

$$\mathbf{m}^\top \mathbf{V}\mathbf{m} = 3, \quad \mathbf{m}^\top (\mathbf{V}\mathbf{h} + \mathbf{u}) = -7, \quad g(\mathbf{h}) = f = 8 \tag{1.2.4.9}$$

From (1.1.11.2), the points of intersection of the parabola with the x-axis satisfy

$$3\kappa^2 - 14\kappa + 8 = 0 \tag{1.2.4.10}$$

which is the same equation as the given quadratic equation for $p = 3$. From (1.1.10.1),

$$\kappa = \frac{14 \pm \sqrt{196 - 96}}{6} = \frac{14 \pm 10}{6} = 4 \text{ or } \frac{2}{3} \tag{1.2.4.11}$$

Therefore, the parabola meets the x-axis at

$$\mathbf{x} = \kappa\mathbf{m} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} \text{ and } \begin{pmatrix} \frac{2}{3} \\ 0 \end{pmatrix} \tag{1.2.4.12}$$

so the roots of the quadratic equation are the intersections of the parabola with the x-axis. The following code generates Fig. 1.2.4.1

codes/root_ratio.py

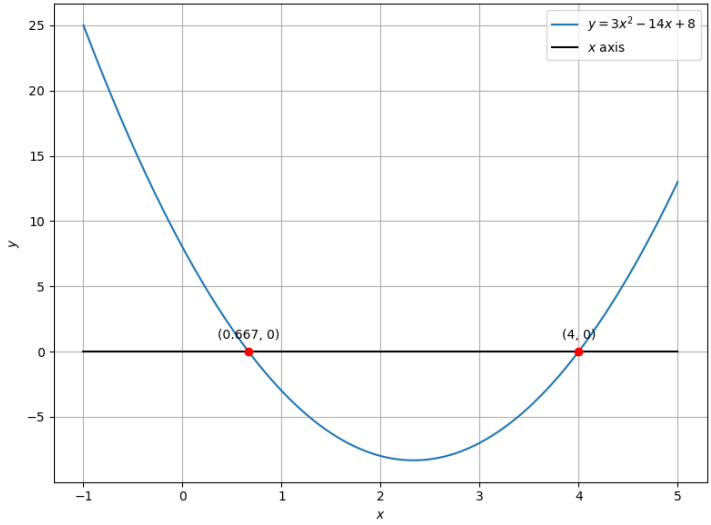


Fig. 1.2.4.1: Parabola $y = 3x^2 - 14x + 8$ meeting the x-axis at $x = \frac{2}{3}$ and $x = 4$

1.2.5 For what value of n, are the nth terms of two A.Ps $63, 65, 67, \ldots$ and $3, 10, 17, \ldots$ equal ? (10, 2017)

Solution: Let the index start from 0, so that $x(n)$ is the $(n+1)^{\text{th}}$ term. From (1.1.14.2), with $a = 63$, $d = 2$ for the first A.P. and $a = 3$, $d = 7$ for the second A.P.,

$$x_1(n) = (63 + 2n)\,u(n), \quad x_2(n) = (3 + 7n)\,u(n) \tag{1.2.5.1}$$

The two sequences intersect when $x_1(n) = x_2(n)$ for $n \geq 0$,

$$63 + 2n = 3 + 7n \tag{1.2.5.2}$$

$$\implies 5n = 60 \tag{1.2.5.3}$$

$$\therefore n = 12 \tag{1.2.5.4}$$

Hence, the stems of the two sequences coincide at $n = 12$, where

$$x_1(12) = 63 + 24 = 87, \quad x_2(12) = 3 + 84 = 87 \tag{1.2.5.5}$$

Since $x(n)$ is the $(n+1)^{\text{th}}$ term, the 13th terms of the two A.Ps are equal. The following code verifies the solution

codes/ap_equal.c

The following code generates Fig. 1.2.5.1

codes/ap_equal.py

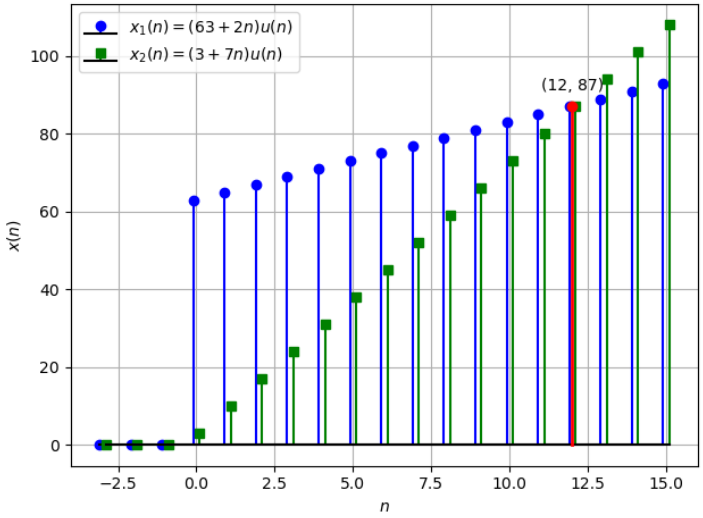


Fig. 1.2.5.1: Stem plot of the two A.Ps, intersecting at $n = 12$

1.2.6 On a straight line passing through the foot of a tower, two points C and D are at distances of 4 m and 16 m from the foot respectively. If the angles of elevation from

C and D of the top of the tower are complementary, then find the height of the tower. (10, 2017)

Solution: Let the foot of the tower be the origin, with the tower along the y-axis and C, D on the positive x-axis. Let h be the height of the tower. Then

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 4 \\ 0 \end{pmatrix}, \quad \mathbf{D} = \begin{pmatrix} 16 \\ 0 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 0 \\ h \end{pmatrix} \tag{1.2.6.1}$$

where $\mathbf{T}$ is the top of the tower. Let θ be the angle of elevation from $\mathbf{C}$, that is, the angle between $\mathbf{T}-\mathbf{C}$ and $\mathbf{O}-\mathbf{C}$. The angle of elevation from $\mathbf{D}$ is then $90^\circ - \theta$. The vectors are

$$\mathbf{T}-\mathbf{C} = \begin{pmatrix} -4 \\ h \end{pmatrix}, \quad \mathbf{O}-\mathbf{C} = \begin{pmatrix} -4 \\ 0 \end{pmatrix}, \quad \mathbf{T}-\mathbf{D} = \begin{pmatrix} -16 \\ h \end{pmatrix}, \quad \mathbf{O}-\mathbf{D} = \begin{pmatrix} -16 \\ 0 \end{pmatrix} \tag{1.2.6.2}$$

From (1.1.4.1),

$$\cos\theta = \frac{(\mathbf{T}-\mathbf{C})^\top (\mathbf{O}-\mathbf{C})}{\|\mathbf{T}-\mathbf{C}\|\,\|\mathbf{O}-\mathbf{C}\|} = \frac{16}{4\sqrt{16+h^2}} = \frac{4}{\sqrt{16+h^2}} \tag{1.2.6.3}$$

$$\cos(90^\circ-\theta) = \frac{(\mathbf{T}-\mathbf{D})^\top (\mathbf{O}-\mathbf{D})}{\|\mathbf{T}-\mathbf{D}\|\,\|\mathbf{O}-\mathbf{D}\|} = \frac{16}{\sqrt{256+h^2}} \tag{1.2.6.4}$$

Since $\cos(90^\circ-\theta) = \sin\theta$, we have $\cos^2\theta + \cos^2(90^\circ-\theta) = 1$,

$$\frac{16}{16+h^2} + \frac{256}{256+h^2} = 1 \tag{1.2.6.5}$$

$$\implies 16\left(256+h^2\right) + 256\left(16+h^2\right) = \left(16+h^2\right)\left(256+h^2\right) \tag{1.2.6.6}$$

$$\implies h^4 = 4096 \tag{1.2.6.7}$$

$$\therefore h = 8 \text{ m} \tag{1.2.6.8}$$

The following code generates Fig. 1.2.6.1

codes/trig_height.py

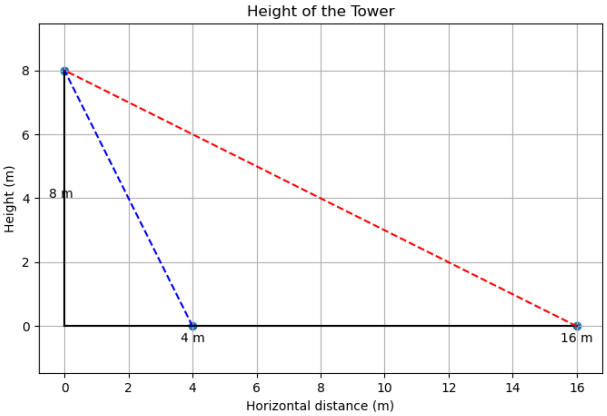


Fig. 1.2.6.1: Tower and the angles of elevation

1.2.7 In what ratio does the point $\left(\frac{24}{11}, y\right)$ divide the line segment joining the points $\mathbf{P}\left(2, -2\right)$ and $\mathbf{Q}\left(3, 7\right)$? Also find the value of y. (10, 2017)

Solution: Let $\mathbf{R} = \begin{pmatrix} \frac{24}{11} \\ y \end{pmatrix}$. Since $\mathbf{R}$ lies on the line segment joining $\mathbf{P}$ and $\mathbf{Q}$, the points $\mathbf{P}$, $\mathbf{Q}$ and $\mathbf{R}$ are collinear. Then

$$\mathbf{Q} - \mathbf{P} = \begin{pmatrix} 1 \\ 9 \end{pmatrix}, \quad \mathbf{R} - \mathbf{P} = \begin{pmatrix} \frac{2}{11} \\ y + 2 \end{pmatrix} \tag{1.2.7.1}$$

From (1.1.9.1),

$$\text{rank}\begin{pmatrix} \mathbf{Q} - \mathbf{P} & \mathbf{R} - \mathbf{P} \end{pmatrix} = \text{rank}\begin{pmatrix} 1 & \frac{2}{11} \\ 9 & y + 2 \end{pmatrix} = 1 \tag{1.2.7.2}$$

Performing the row operation $R_2 \leftarrow R_2 - 9R_1$,

$$\begin{pmatrix} 1 & \frac{2}{11} \\ 9 & y + 2 \end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2 - 9R_1} \begin{pmatrix} 1 & \frac{2}{11} \\ 0 & y + 2 - \frac{18}{11} \end{pmatrix} \tag{1.2.7.3}$$

For the rank to be 1, the second row must vanish,

$$y + 2 - \frac{18}{11} = 0 \implies y = -\frac{4}{11} \tag{1.2.7.4}$$

The second column is also $\frac{2}{11}$ times the first column, so

$$\mathbf{R} - \mathbf{P} = \frac{2}{11}\left(\mathbf{Q} - \mathbf{P}\right) \tag{1.2.7.5}$$

From (1.1.8.1) with $\lambda = \frac{2}{11}$,

$$\mathbf{PR} : \mathbf{RQ} = \lambda : (1 - \lambda) = \frac{2}{11} : \frac{9}{11} = 2 : 9 \tag{1.2.7.6}$$

Thus, the ratio is $2 : 9$ and $y = -\frac{4}{11}$. The following code verifies the solution

codes/section.c

1.2.8 How many terms of an A.P. $9, 17, 25, \ldots$ must be taken to give a sum of 636 ? (10, 2017)

Solution: Let $x(n)$ be the $(n+1)^{\text{th}}$ term of the A.P. $9, 17, 25, \ldots$, where $n = 0, 1, 2, \ldots$ Then

$$x(n) = (9 + 8n)\, u(n) \tag{1.2.8.1}$$

The sum of the first $n+1$ terms is

$$y(n) = \sum_{k=0}^{n} x(k) = x(n) * u(n) \tag{1.2.8.2}$$

From (1.1.15.2), for $|z| > 1$,

$$Y(z) = \frac{X(z)}{1 - z^{-1}} \tag{1.2.8.3}$$

Using (1.1.16.1),

$$X(z) = \frac{9}{1 - z^{-1}} + \frac{8z^{-1}}{(1 - z^{-1})^2} = \frac{9 - z^{-1}}{(1 - z^{-1})^2} \tag{1.2.8.4}$$

$$\implies Y(z) = \frac{9 - z^{-1}}{(1 - z^{-1})^3} = \frac{9}{(1 - z^{-1})^3} - \frac{z^{-1}}{(1 - z^{-1})^3} \tag{1.2.8.5}$$

Taking the inverse Z-transform using (1.1.16.1) and (1.1.15.2),

$$y(n) = 9 \cdot \frac{(n+1)(n+2)}{2} u(n) - \frac{n(n+1)}{2} u(n-1) \tag{1.2.8.6}$$

The second term is zero at $n = 0$, so $u(n-1)$ can be replaced by $u(n)$,

$$y(n) = \frac{(n+1)(9n + 18 - n)}{2} u(n) = (n+1)(4n+9)\, u(n) \tag{1.2.8.7}$$

Equating the sum to 636,

$$(n+1)(4n+9) = 636 \tag{1.2.8.8}$$

$$\implies 4n^2 + 13n - 627 = 0 \tag{1.2.8.9}$$

From (1.1.10.1),

$$n = \frac{-13 \pm \sqrt{169 + 10032}}{8} = \frac{-13 \pm 101}{8} \tag{1.2.8.10}$$

Since $n \geq 0$,

$$n = 11 \tag{1.2.8.11}$$

Hence, the number of terms is $n + 1 = 12$. The following code verifies the solution

codes/ap_sum.c

1.2.9 If the points $\mathbf{A}\left(k+1, 2k\right)$, $\mathbf{B}\left(3k, 2k+3\right)$ and $\mathbf{C}\left(5k-1, 5k\right)$ are collinear, then find the value of k. (10, 2017)

Solution: Let

$$\mathbf{B}-\mathbf{A}=\begin{pmatrix}2k-1\\3\end{pmatrix}, \quad \mathbf{C}-\mathbf{A}=\begin{pmatrix}4k-2\\3k\end{pmatrix} \tag{1.2.9.1}$$

From (1.1.9.1), the points are collinear if and only if

$$\text{rank}\begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A}\end{pmatrix}=\text{rank}\begin{pmatrix}2k-1 & 4k-2\\3 & 3k\end{pmatrix}=1 \tag{1.2.9.2}$$

Performing the row operations $R_1 \leftrightarrow R_2$ and $R_1 \leftarrow \frac{R_1}{3}$,

$$\begin{pmatrix}2k-1 & 4k-2\\3 & 3k\end{pmatrix} \xleftrightarrow[R_1 \leftarrow \frac{R_1}{3}]{R_1 \leftrightarrow R_2} \begin{pmatrix}1 & k\\2k-1 & 4k-2\end{pmatrix} \tag{1.2.9.3}$$

Performing the row operation $R_2 \leftarrow R_2-(2k-1)R_1$,

$$\begin{pmatrix}1 & k\\2k-1 & 4k-2\end{pmatrix} \xleftrightarrow{R_2 \leftarrow R_2-(2k-1)R_1} \begin{pmatrix}1 & k\\0 & -2k^2+5k-2\end{pmatrix} \tag{1.2.9.4}$$

The first row is nonzero, so for the rank to be 1, the second row must vanish,

$$-2k^2+5k-2=0 \tag{1.2.9.5}$$

$$\implies 2k^2-5k+2=0 \tag{1.2.9.6}$$

$$\implies (2k-1)(k-2)=0 \tag{1.2.9.7}$$

Hence,

$$k=\frac{1}{2} \quad \text{or} \quad k=2 \tag{1.2.9.8}$$

1.2.10 Two different dice are thrown together. Find the probability that the numbers obtained have

a) even sum, and
b) even product.

(10, 2017)

Solution: Let X_1 and X_2 represent the scores on the two dice, and let $X=X_1+X_2$ denote their sum.

a) From the triangular PMF in (1.1.18.3), the sum is even for $n \in \{2,4,6,8,10,12\}$. Hence,

$$\Pr\left(X \text{ is even}\right)=\sum_{n\in\{2,4,6,8,10,12\}} p_X\left(n\right) \tag{1.2.10.1}$$

$$=\frac{1+3+5+5+3+1}{36}=\frac{1}{2} \tag{1.2.10.2}$$

b) The product X_1X_2 is odd if and only if both X_1 and X_2 are odd. From (1.1.18.1),

$$\Pr\left(X_1 \text{ is odd}\right)=p_{X_1}\left(1\right)+p_{X_1}\left(3\right)+p_{X_1}\left(5\right)=\frac{3}{6}=\frac{1}{2} \tag{1.2.10.3}$$

and similarly $\Pr(X_2 \text{ is odd}) = \frac{1}{2}$. Since X_1 and X_2 are independent,

$$\Pr(X_1X_2 \text{ is odd}) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4} \tag{1.2.10.4}$$

Using the complement in (1.1.17.1),

$$\Pr(X_1X_2 \text{ is even}) = 1 - \frac{1}{4} = \frac{3}{4} \tag{1.2.10.5}$$

1.2.11 In the given figure, XY and $X'Y'$ are two parallel tangents to a circle with centre O and another tangent AB with point of contact C, is intersecting XY at A and $X'Y'$ at B. Prove that $\angle AOB = 90°$. (10, 2017)

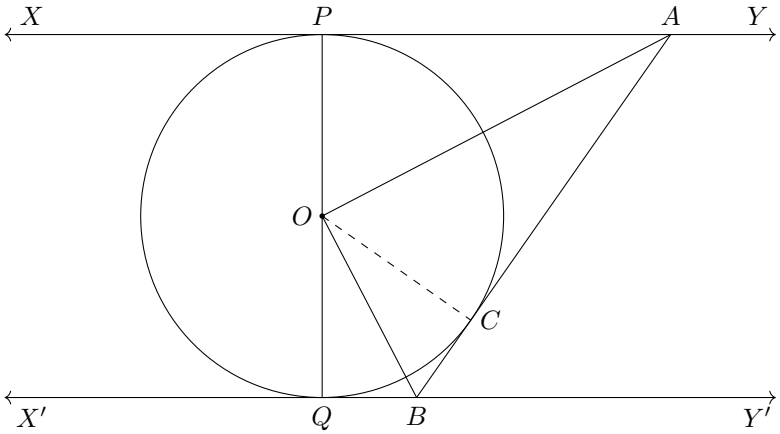


Fig. 1.2.11.1

Solution: In Fig. 1.2.11.1, let r be the radius of the circle, and let P, Q and C be the points of contact of XY, $X'Y'$ and AB. Take **O** as the origin with $XY \parallel X'Y'$ along the x-direction. Since the tangents at P and Q are parallel, from (1.1.1.1), P, O and Q are collinear,

$$\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix}, \quad \mathbf{P} = \begin{pmatrix}0\\r\end{pmatrix}, \quad \mathbf{Q} = \begin{pmatrix}0\\-r\end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix}c_1\\c_2\end{pmatrix}, \quad c_1^2 + c_2^2 = r^2 \tag{1.2.11.1}$$

The points **A** and **B** lie on the tangents XY and $X'Y'$ respectively, so

$$\mathbf{A} = \begin{pmatrix}a\\r\end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix}b\\-r\end{pmatrix} \tag{1.2.11.2}$$

AB touches the circle at **C** and **A**, **B** lie on this tangent. From (1.1.1.1) with **O** as the origin,

$$(\mathbf{A}-\mathbf{C})^\top \mathbf{C} = 0 \implies \mathbf{A}^\top\mathbf{C} = \|\mathbf{C}\|^2 = r^2 \tag{1.2.11.3}$$

$$(\mathbf{B}-\mathbf{C})^\top \mathbf{C} = 0 \implies \mathbf{B}^\top\mathbf{C} = r^2 \tag{1.2.11.4}$$

Substituting the coordinates,

$$ac_1 + rc_2 = r^2 \implies ac_1 = r\left(r - c_2\right) \tag{1.2.11.5}$$

$$bc_1 - rc_2 = r^2 \implies bc_1 = r\left(r + c_2\right) \tag{1.2.11.6}$$

Multiplying the two equations,

$$ab\,c_1^2 = r^2\left(r^2 - c_2^2\right) = r^2c_1^2 \tag{1.2.11.7}$$

Since $c_1 \neq 0$ (otherwise AB would coincide with XY or $X'Y'$),

$$ab = r^2 \tag{1.2.11.8}$$

From (1.1.4.1),

$$\cos\angle AOB = \frac{\mathbf{A}^\top\mathbf{B}}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = \frac{ab - r^2}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = 0 \tag{1.2.11.9}$$

Hence,

$$\angle AOB = 90^\circ \tag{1.2.11.10}$$

To verify numerically, take a circle with centre $\mathbf{O}$ and radius $r = 10$, and the point of contact $\mathbf{C}$ of AB,

$$\mathbf{O} = \begin{pmatrix}0\\0\end{pmatrix},\quad \mathbf{P} = \begin{pmatrix}0\\10\end{pmatrix},\quad \mathbf{Q} = \begin{pmatrix}0\\-10\end{pmatrix},\quad \mathbf{C} = \begin{pmatrix}8\\-6\end{pmatrix} \tag{1.2.11.11}$$

Here $\|\mathbf{C}\|^2 = 64 + 36 = 100 = r^2$, so $\mathbf{C}$ lies on the circle. The points $\mathbf{A}$ and $\mathbf{B}$ lie on the tangents $y = 10$ and $y = -10$ respectively, so let

$$\mathbf{A} = \begin{pmatrix}a\\10\end{pmatrix},\quad \mathbf{B} = \begin{pmatrix}b\\-10\end{pmatrix} \tag{1.2.11.12}$$

From (1.1.1.1), $\mathbf{A}^\top\mathbf{C} = \mathbf{B}^\top\mathbf{C} = r^2 = 100$,

$$8a - 60 = 100 \implies a = 20 \tag{1.2.11.13}$$

$$8b + 60 = 100 \implies b = 5 \tag{1.2.11.14}$$

Hence,

$$\mathbf{A} = \begin{pmatrix}20\\10\end{pmatrix},\quad \mathbf{B} = \begin{pmatrix}5\\-10\end{pmatrix} \tag{1.2.11.15}$$

Then $ab = 100 = r^2$, and the tangent AB is perpendicular to the radius at $\mathbf{C}$,

$$\left(\mathbf{A} - \mathbf{C}\right)^\top\mathbf{C} = \begin{pmatrix}12 & 16\end{pmatrix}\begin{pmatrix}8\\-6\end{pmatrix} = 0 \tag{1.2.11.16}$$

$$\left(\mathbf{B} - \mathbf{C}\right)^\top\mathbf{C} = \begin{pmatrix}-3 & -4\end{pmatrix}\begin{pmatrix}8\\-6\end{pmatrix} = 0 \tag{1.2.11.17}$$

The tangent lengths from $\mathbf{A}$ and $\mathbf{B}$ are equal, from (1.1.2.1),

$$\|\mathbf{A}-\mathbf{P}\| = \left\|\begin{pmatrix}20\\0\end{pmatrix}\right\| = 20 = \left\|\begin{pmatrix}12\\16\end{pmatrix}\right\| = \|\mathbf{A}-\mathbf{C}\| \tag{1.2.11.18}$$

$$\|\mathbf{B}-\mathbf{Q}\| = \left\|\begin{pmatrix}5\\0\end{pmatrix}\right\| = 5 = \left\|\begin{pmatrix}-3\\-4\end{pmatrix}\right\| = \|\mathbf{B}-\mathbf{C}\| \tag{1.2.11.19}$$

Finally, from (1.1.4.1),

$$\cos\angle AOB = \frac{\mathbf{A}^\top\mathbf{B}}{\|\mathbf{A}\|\,\|\mathbf{B}\|} = \frac{100-100}{\sqrt{500}\sqrt{125}} = 0 \tag{1.2.11.20}$$

$$\implies \angle AOB = 90^\circ \tag{1.2.11.21}$$

which verifies the result. The following code generates Fig. 1.2.11.2

codes/tangent_angle.py

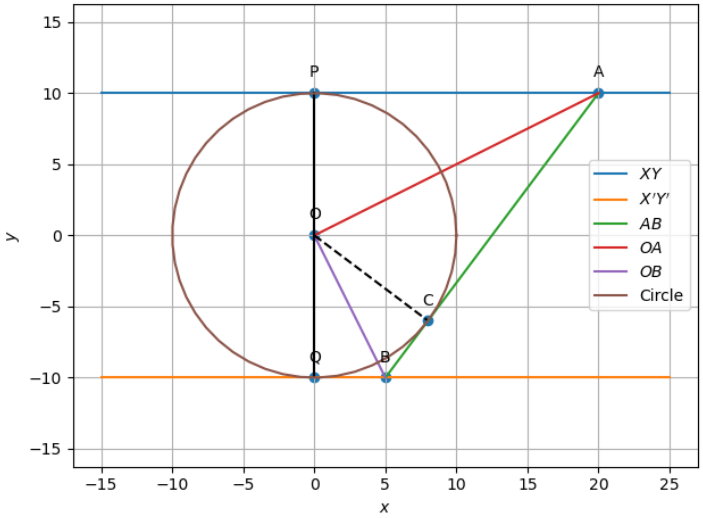


Fig. 1.2.11.2: Tangents from $\mathbf{h}$ to the circle, with the points of contact $\mathbf{T}_1$ and $\mathbf{T}_2$

1.2.12 Prove that the lengths of two tangents drawn from an external point to a circle are equal. (10, 2017)

Solution: Let $\mathbf{P}$ be an external point of a circle with centre $\mathbf{O}$ and radius a. Let the tangents from $\mathbf{P}$ touch the circle at $\mathbf{T}_1$ and $\mathbf{T}_2$. From (1.1.1.1), the triangles $\mathbf{OT}_1\mathbf{P}$ and $\mathbf{OT}_2\mathbf{P}$ are right angled at $\mathbf{T}_1$ and $\mathbf{T}_2$. By Pythagoras' theorem,

$$\|\mathbf{P}-\mathbf{T}_1\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - \|\mathbf{T}_1-\mathbf{O}\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - a^2 \tag{1.2.12.1}$$

$$\|\mathbf{P}-\mathbf{T}_2\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - \|\mathbf{T}_2-\mathbf{O}\|^2 = \|\mathbf{P}-\mathbf{O}\|^2 - a^2 \tag{1.2.12.2}$$

Therefore,

$$\|\mathbf{P}-\mathbf{T}_1\| = \|\mathbf{P}-\mathbf{T}_2\| \tag{1.2.12.3}$$

To verify numerically, take a circle with centre $\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and radius $r = 5$, and the external point $\mathbf{h} = \begin{pmatrix} 5 \\ 10 \end{pmatrix}$. From (1.1.12.1),

$$\mathbf{V} = \mathbf{I}, \quad \mathbf{u} = -\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad f = \|\mathbf{O}\|^2 - r^2 = -25 \tag{1.2.12.4}$$

and

$$g(\mathbf{h}) = \|\mathbf{h}\|^2 + 2\mathbf{u}^\top \mathbf{h} + f = 125 - 25 = 100 > 0 \tag{1.2.12.5}$$

so $\mathbf{h}$ is an external point. From (1.1.13.1),

$$\mathbf{\Sigma} = \begin{pmatrix} 5 \\ 10 \end{pmatrix} \begin{pmatrix} 5 & 10 \end{pmatrix} - 100\mathbf{I} = \begin{pmatrix} -75 & 50 \\ 50 & 0 \end{pmatrix} \tag{1.2.12.6}$$

The eigenvalues are the roots of

$$\left|\lambda\mathbf{I} - \mathbf{\Sigma}\right| \Rightarrow \lambda^2 + 75\lambda - 2500 = (\lambda - 25)(\lambda + 100) = 0 \tag{1.2.12.7}$$

so $\lambda_1 = 25$ and $\lambda_2 = -100$. For $\lambda_1 = 25$, using row reduction,

$$\mathbf{\Sigma} - 25\mathbf{I} = \begin{pmatrix} -100 & 50 \\ 50 & -25 \end{pmatrix} \xleftarrow[R_2 \leftarrow R_2 - 50R_1]{R_1 \leftarrow -\frac{R_1}{100}} \begin{pmatrix} 1 & -\frac{1}{2} \\ 0 & 0 \end{pmatrix} \implies \mathbf{p}_1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \tag{1.2.12.8}$$

For $\lambda_2 = -100$,

$$\mathbf{\Sigma} + 100\mathbf{I} = \begin{pmatrix} 25 & 50 \\ 50 & 100 \end{pmatrix} \xleftarrow[R_2 \leftarrow R_2 - 50R_1]{R_1 \leftarrow \frac{R_1}{25}} \begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix} \implies \mathbf{p}_2 = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \tag{1.2.12.9}$$

Hence,

$$\mathbf{D} = \begin{pmatrix} 25 & 0 \\ 0 & -100 \end{pmatrix}, \quad \mathbf{P} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} \tag{1.2.12.10}$$

Here $\mathbf{P}^\top \mathbf{P} = \mathbf{I}$, so $\mathbf{P}^\top = \mathbf{P}^{-1}$, and one can verify that $\mathbf{P}^\top \mathbf{\Sigma} \mathbf{P} = \mathbf{D}$. From (1.1.13.3), the direction vectors of the tangents are

$$\mathbf{m} = \mathbf{P} \begin{pmatrix} \sqrt{100} \\ \pm\sqrt{25} \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 1 & 2 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 10 \\ \pm 5 \end{pmatrix} = \frac{1}{\sqrt{5}} \begin{pmatrix} 20 \\ 15 \end{pmatrix} \text{ or } \frac{1}{\sqrt{5}} \begin{pmatrix} 0 \\ 25 \end{pmatrix} \tag{1.2.12.11}$$

Only the direction of $\mathbf{m}$ matters, since $\mu\mathbf{m}$ does not change on scaling $\mathbf{m}$. So take

$$\mathbf{m}_1 = \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \quad \mathbf{m}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \tag{1.2.12.12}$$

From (1.1.13.4), the points of contact $\mathbf{T}_i = \mathbf{h} + \mu_i \mathbf{m}_i$ are given by

$$\mu_1 = -\frac{\mathbf{m}_1^\top (\mathbf{h}+\mathbf{u})}{\mathbf{m}_1^\top \mathbf{m}_1} = -\frac{50}{25} = -2, \quad \mathbf{T}_1 = \begin{pmatrix} 5 \\ 10 \end{pmatrix} - 2\begin{pmatrix} 4 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 4 \end{pmatrix} \tag{1.2.12.13}$$

$$\mu_2 = -\frac{\mathbf{m}_2^\top (\mathbf{h}+\mathbf{u})}{\mathbf{m}_2^\top \mathbf{m}_2} = -\frac{10}{1} = -10, \quad \mathbf{T}_2 = \begin{pmatrix} 5 \\ 10 \end{pmatrix} - 10\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix} \tag{1.2.12.14}$$

Both points lie on the circle, since $\|\mathbf{T}_1\|^2 = 9 + 16 = 25$ and $\|\mathbf{T}_2\|^2 = 25$. Also, from (1.1.1.1),

$$(\mathbf{h} - \mathbf{T}_1)^\top (\mathbf{T}_1 - \mathbf{O}) = \begin{pmatrix} 8 & 6 \end{pmatrix} \begin{pmatrix} -3 \\ 4 \end{pmatrix} = 0 \tag{1.2.12.15}$$

$$(\mathbf{h} - \mathbf{T}_2)^\top (\mathbf{T}_2 - \mathbf{O}) = \begin{pmatrix} 0 & 10 \end{pmatrix} \begin{pmatrix} 5 \\ 0 \end{pmatrix} = 0 \tag{1.2.12.16}$$

so the tangents are perpendicular to the radii. The lengths of the tangents are

$$\|\mathbf{h} - \mathbf{T}_1\| = \left\| \begin{pmatrix} 8 \\ 6 \end{pmatrix} \right\| = 10 \tag{1.2.12.17}$$

$$\|\mathbf{h} - \mathbf{T}_2\| = \left\| \begin{pmatrix} 0 \\ 10 \end{pmatrix} \right\| = 10 \tag{1.2.12.18}$$

which are equal and agree with $\sqrt{\|\mathbf{h} - \mathbf{O}\|^2 - r^2} = \sqrt{125 - 25} = 10$ from (1.1.2.1). The following code generates Fig. 1.2.12.1

codes/tangent_equal.py

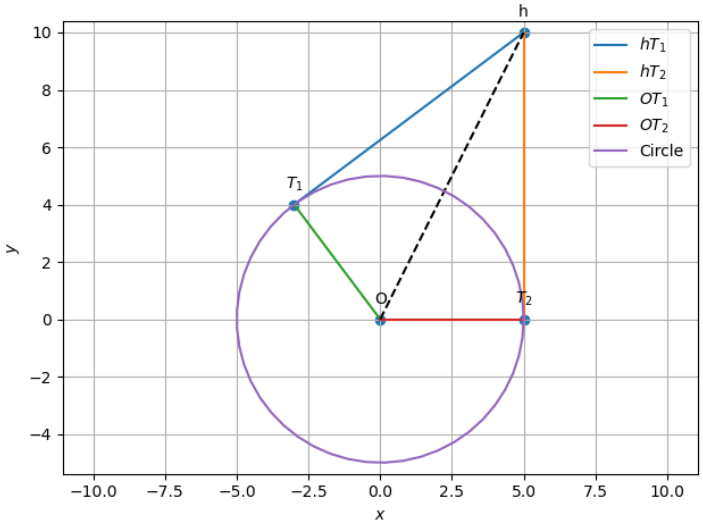


Fig. 1.2.12.1: Equal tangent lengths $\|\mathbf{P}-\mathbf{T}_1\| = \|\mathbf{P}-\mathbf{T}_2\|$

1.2.13 Solve for x :

$$\frac{x-1}{2x+1}+\frac{2x+1}{x-1}=2, \quad \text{where } x \neq -\frac{1}{2}, 1$$

(10, 2017)

Solution: Multiplying by $(2x+1)(x-1)$,

$$(x-1)^2+(2x+1)^2 = 2(2x+1)(x-1) \tag{1.2.13.1}$$

$$\implies 5x^2+2x+2 = 4x^2-2x-2 \tag{1.2.13.2}$$

$$\implies x^2+4x+4 = 0 \tag{1.2.13.3}$$

$$\implies (x+2)^2 = 0 \tag{1.2.13.4}$$

Here $q^2-4pr = 16-16 = 0$, so from (1.1.10.2) the roots are equal,

$$x = -2 \tag{1.2.13.5}$$

which satisfies $x \neq -\frac{1}{2}, 1$. The following code generates Fig. 1.2.13.1

codes/equal_roots.py

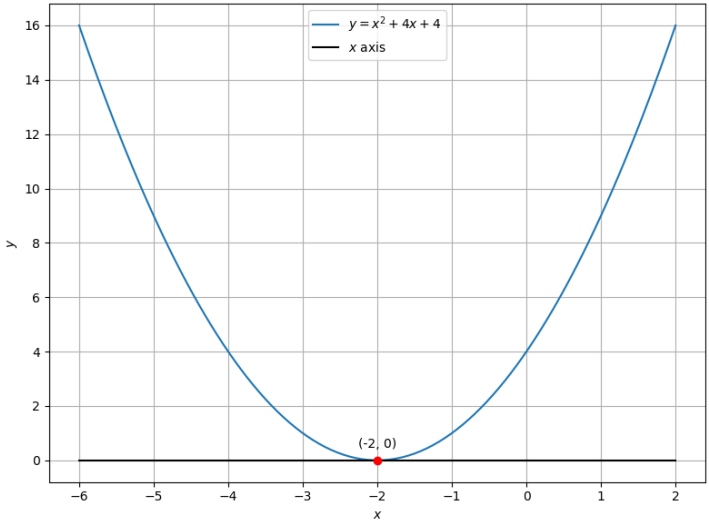


Fig. 1.2.13.1: Parabola touching the x-axis at $x = -2$

1.2.14 A takes 6 days less than B to do a work. If both A and B working together can do it in 4 days, how many days will B take to finish it ? (10, 2017)

Solution: Let B take x days to finish the work. Then A takes $x - 6$ days. From (1.1.19.1),

$$\frac{1}{x} + \frac{1}{x-6} = \frac{1}{4} \tag{1.2.14.1}$$

$$\implies 4(2x - 6) = x(x - 6) \tag{1.2.14.2}$$

$$\implies x^2 - 14x + 24 = 0 \tag{1.2.14.3}$$

From (1.1.10.1),

$$x = \frac{14 \pm \sqrt{196 - 96}}{2} = \frac{14 \pm 10}{2} = 12 \text{ or } 2 \tag{1.2.14.4}$$

For $x = 2$, A would take -4 days, which is not possible. Therefore, B takes 12 days. The following code verifies the solution

codes/work_time.c

The following code generates Fig. 1.2.14.1

codes/work_time.py

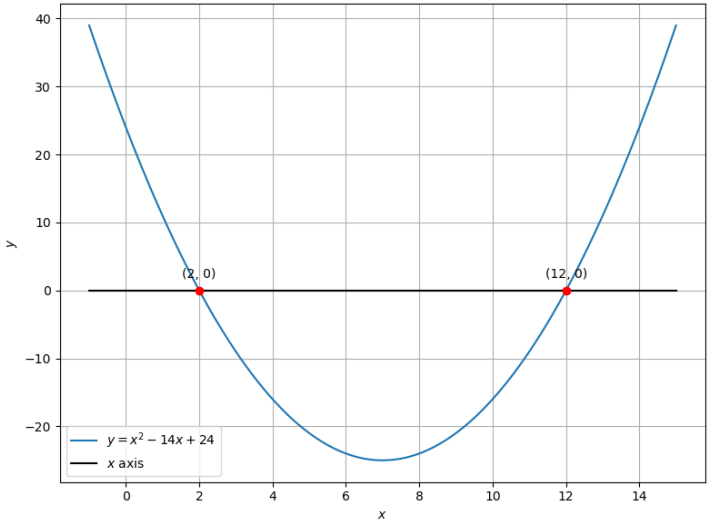


Fig. 1.2.14.1: Work done by A and B

1.2.15 From the top of a tower, 100 m high, a man observes two cars on the opposite sides of the tower and in same straight line with its base, with angles of depression $30°$ and $45°$. Find the distance between the cars. [Take $\sqrt{3} = 1.732$] (10, 2017)

Solution: Let the foot of the tower be the origin, with the tower along the y-axis and the cars on the x-axis on opposite sides of the tower. Let the cars be at distances d_1 and d_2 from the foot, with angles of depression $45°$ and $30°$ respectively,

$$\mathbf{O} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \quad \mathbf{T} = \begin{pmatrix} 0 \\ 100 \end{pmatrix}, \quad \mathbf{C}_1 = \begin{pmatrix} d_1 \\ 0 \end{pmatrix}, \quad \mathbf{C}_2 = \begin{pmatrix} -d_2 \\ 0 \end{pmatrix} \tag{1.2.15.1}$$

where $\mathbf{T}$ is the top of the tower. The vectors are

$$\mathbf{T} - \mathbf{C}_1 = \begin{pmatrix} -d_1 \\ 100 \end{pmatrix}, \quad \mathbf{O} - \mathbf{C}_1 = \begin{pmatrix} -d_1 \\ 0 \end{pmatrix}, \quad \mathbf{T} - \mathbf{C}_2 = \begin{pmatrix} d_2 \\ 100 \end{pmatrix}, \quad \mathbf{O} - \mathbf{C}_2 = \begin{pmatrix} d_2 \\ 0 \end{pmatrix} \tag{1.2.15.2}$$

From (1.1.20.1),

$$\cos 45° = \frac{(\mathbf{T} - \mathbf{C}_1)^\top (\mathbf{O} - \mathbf{C}_1)}{\|\mathbf{T} - \mathbf{C}_1\| \, \|\mathbf{O} - \mathbf{C}_1\|} = \frac{d_1}{\sqrt{d_1^2 + 10000}} \tag{1.2.15.3}$$

$$\cos 30° = \frac{(\mathbf{T} - \mathbf{C}_2)^\top (\mathbf{O} - \mathbf{C}_2)}{\|\mathbf{T} - \mathbf{C}_2\| \, \|\mathbf{O} - \mathbf{C}_2\|} = \frac{d_2}{\sqrt{d_2^2 + 10000}} \tag{1.2.15.4}$$

Squaring both equations,

$$\frac{1}{2} = \frac{d_1^2}{d_1^2 + 10000} \implies d_1^2 = 10000 \implies d_1 = 100 \text{ m} \tag{1.2.15.5}$$

$$\frac{3}{4} = \frac{d_2^2}{d_2^2 + 10000} \implies d_2^2 = 30000 \implies d_2 = 100\sqrt{3} \text{ m} \tag{1.2.15.6}$$

The distance between the cars is

$$\|\mathbf{C}_1 - \mathbf{C}_2\| = \left\| \begin{pmatrix} d_1 + d_2 \\ 0 \end{pmatrix} \right\| = d_1 + d_2 = 100\left(1 + \sqrt{3}\right) = 100\,(1 + 1.732) = 273.2 \text{ m} \tag{1.2.15.7}$$

The following code generates Fig. 1.2.15.1

codes/tower_cars.py

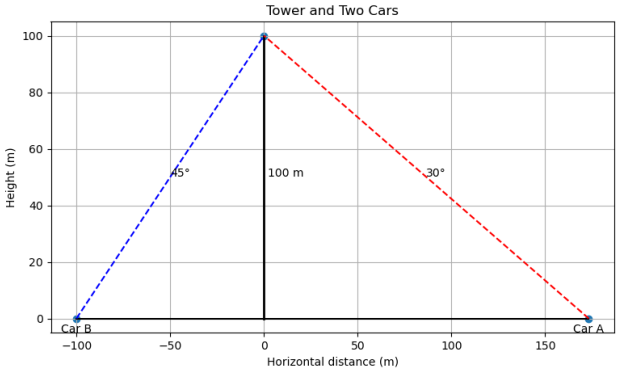


Fig. 1.2.15.1: Tower and the two cars

2. 12

A. Formulae

2.1.1 The normal form of the equation of a line is

$$\mathbf{n}^\top \mathbf{x} = c \tag{2.1.1.1}$$

The same form represents a plane when $\mathbf{x}$ and $\mathbf{n}$ are 3-dimensional vectors.

2.1.2 The distance between the parallel planes $\mathbf{n}^\top \mathbf{x} = c_1$ and $\mathbf{n}^\top \mathbf{x} = c_2$ is

$$d = \frac{|c_2 - c_1|}{\|\mathbf{n}\|} \tag{2.1.2.1}$$

Solution: The point $\mathbf{x}_1 = \frac{c_1}{\|\mathbf{n}\|^2}\mathbf{n}$ lies on the first plane, from (2.1.1.1). Moving along $\mathbf{n}$, $\mathbf{x}_1 + t\mathbf{n}$ lies on the second plane if

$$\mathbf{n}^\top (\mathbf{x}_1 + t\mathbf{n}) = c_2 \implies c_1 + t\|\mathbf{n}\|^2 = c_2 \implies t = \frac{c_2 - c_1}{\|\mathbf{n}\|^2} \tag{2.1.2.2}$$

The distance is $|t|\,\|\mathbf{n}\|$, yielding (2.1.2.1).

2.1.3 For a square matrix $\mathbf{A}$ of order n,

$$\mathbf{A}\,\mathrm{adj}\,(\mathbf{A}) = \left|\mathbf{A}\right|\mathbf{I} \tag{2.1.3.1}$$

2.1.4 (Trace of a Matrix) Let $\mathbf{A} \in \mathbb{R}^{n\times n}$ be a square matrix with eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Then,

$$\sum_{i=1}^{n} \lambda_i = \sum_{i=1}^{n} a_{ii} = \mathrm{tr}\,(\mathbf{A}) \tag{2.1.4.1}$$

2.1.5 The probability mass function (PMF) of the outcome X of a fair die is

$$p_X(n) = \begin{cases} \frac{1}{6}, & 1 \le n \le 6 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.5.1}$$

The probability of an event E is

$$\Pr(E) = \sum_{n\in E} p_X(n) \tag{2.1.5.2}$$

2.1.6 The events A and B are independent if and only if

$$\Pr(AB) = \Pr(A)\Pr(B) \tag{2.1.6.1}$$

2.1.7 (Continuity) A real-valued function $f(x)$ is continuous at $x = x_0$ if and only if

$$\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x) = f(x_0) \tag{2.1.7.1}$$

2.1.8 (Differentiability) A function $f(x)$ is differentiable at $x = x_0$ if

$$f'\left(x_0^-\right) = \lim_{h\to 0^-} \frac{f(x_0+h) - f(x_0)}{h} = \lim_{h\to 0^+} \frac{f(x_0+h) - f(x_0)}{h} = f'\left(x_0^+\right) \tag{2.1.8.1}$$

2.1.9 If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$.
Solution:

$$\lim_{x\to x_0} \left[f(x) - f(x_0)\right] = \lim_{x\to x_0} \left[\frac{f(x) - f(x_0)}{x - x_0}(x - x_0)\right] = f'(x_0)\cdot 0 = 0 \tag{2.1.9.1}$$

2.1.10 (Piecewise Boundary Conditions at $x = a$) Consider a function defined by differen-

tiable branches $f_1(x)$ and $f_2(x)$,

$$f(x) = \begin{cases} f_1(x), & x \le a \\ f_2(x), & x > a \end{cases} \tag{2.1.10.1}$$

For $f(x)$ to be continuous at $x = a$,

$$\lim_{x \to a^-} f_1(x) = \lim_{x \to a^+} f_2(x) \implies f_1(a) = f_2(a) \tag{2.1.10.2}$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives must coincide,

$$f_1'(a) = f_2'(a) \tag{2.1.10.3}$$

2.1.11 (Newton-Raphson Method) For a real-valued differentiable function $g(x)$, the root of $g(x) = 0$ is iteratively approximated using the first-order Taylor series expansion about the current estimate x_n, which gives the difference equation

$$x_{n+1} = x_n - \frac{g(x_n)}{g'(x_n)} \tag{2.1.11.1}$$

Solution: The first-order Taylor series of $g(x)$ about x_n is

$$g(x) \approx g(x_n) + g'(x_n)(x - x_n) \tag{2.1.11.2}$$

Equating the right side to zero and calling the solution x_{n+1} yields (2.1.11.1). If $x_n \to x^*$, then $x^* = x^* - \frac{g(x^*)}{g'(x^*)}$, so $g(x^*) = 0$, that is, the limit is a root of $g(x)$.

2.1.12 For a quadratic equation $g(x) = ax^2 + bx + c = 0$, we have $g'(x) = 2ax + b$, and the difference equation in (2.1.11.1) becomes

$$x_{n+1} = x_n - \frac{ax_n^2 + bx_n + c}{2ax_n + b} = \frac{ax_n^2 - c}{2ax_n + b} \tag{2.1.12.1}$$

which converges to the root nearest to x_0, provided $2ax_0 + b \neq 0$.

2.1.13 (Rolle's Theorem) If $f(x)$ is continuous on $[a, b]$, differentiable on (a, b) and $f(a) = f(b)$, then there exists $c \in (a, b)$ such that

$$f'(c) = 0 \tag{2.1.13.1}$$

2.1.14 If $f'(x) > 0$ for all x in an interval, then $f(x)$ is strictly increasing on that interval.

2.1.15 (Trapezoidal Rule) Divide $[a, b]$ into N equal sub-intervals of width $h = \frac{b-a}{N}$, with $x_n = a + nh$. Approximating $f(x)$ by a straight line on each sub-interval, the integral is computed by the difference equation

$$I_{n+1} = I_n + \frac{h}{2}\left(f(x_n) + f(x_{n+1})\right), \quad I_0 = 0 \tag{2.1.15.1}$$

and

$$\int_a^b f(x)\, dx \approx I_N \tag{2.1.15.2}$$

Solution: On $[x_n, x_{n+1}]$, the area under the straight line joining $(x_n, f(x_n))$ and

$(x_{n+1}, f(x_{n+1}))$ is a trapezoid,

$$\int_{x_n}^{x_{n+1}} f(x)\,dx \approx \frac{x_{n+1}-x_n}{2}\left(f(x_n)+f(x_{n+1})\right)$$
$$= \frac{h}{2}\left(f(x_n)+f(x_{n+1})\right) \tag{2.1.15.3}$$

Adding this area to the integral $I_n = \int_a^{x_n} f(x)\,dx$ obtained so far yields (2.1.15.1). For a twice differentiable $f(x)$, the error is

$$\int_a^b f(x)\,dx - I_N = -\frac{(b-a)\,h^2}{12} f''(\xi), \quad \xi \in (a,b) \tag{2.1.15.4}$$

so it decreases as h^2.

2.1.16 For a continuous function $f(x)$,

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx \tag{2.1.16.1}$$

Solution: Substituting $x = a - t$, $dx = -dt$, the limits $x = 0, a$ become $t = a, 0$,

$$\int_0^a f(x)\,dx = -\int_a^0 f(a-t)\,dt = \int_0^a f(a-t)\,dt \tag{2.1.16.2}$$

2.1.17 Trigonometric identities,

$$\tan(\pi - x) = -\tan x, \quad \sec(\pi - x) = -\sec x \tag{2.1.17.1}$$

$$\frac{\tan x}{\sec x + \tan x} = \frac{\sin x}{1+\sin x}, \quad \frac{1}{1+\sin x} = \sec^2 x - \sec x \tan x \tag{2.1.17.2}$$

Solution: Multiplying the numerator and the denominator by $\cos x$ gives the first identity. For the second,

$$\frac{1}{1+\sin x} = \frac{1-\sin x}{1-\sin^2 x} = \frac{1-\sin x}{\cos^2 x} \tag{2.1.17.3}$$

which equals $\sec^2 x - \sec x \tan x$.

2.1.18 The derivatives of $\tan x$ and $\sec x$ are

$$\frac{d}{dx}\tan x = \sec^2 x, \quad \frac{d}{dx}\sec x = \sec x \tan x \tag{2.1.18.1}$$

2.1.19 The absolute value is defined as

$$|x-a| = \begin{cases} x-a, & x \geq a \\ a-x, & x < a \end{cases} \tag{2.1.19.1}$$

2.1.20 Two vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if

$$\mathbf{u}^\top \mathbf{v} = 0 \tag{2.1.20.1}$$

2.1.21 The area of a triangle with two sides $\mathbf{u}$ and $\mathbf{v}$ meeting at a vertex is

$$A = \frac{1}{2} \|\mathbf{u} \times \mathbf{v}\| \tag{2.1.21.1}$$

where, for $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$,

$$\mathbf{u} \times \mathbf{v} = \begin{pmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{pmatrix} \tag{2.1.21.2}$$

2.1.22 The mean and the variance of a discrete random variable X with PMF $p_X(n)$ are

$$E(X) = \sum_n n\, p_X(n), \quad \text{Var}(X) = \sum_n n^2\, p_X(n) - (E(X))^2 \tag{2.1.22.1}$$

2.1.23 (Sum of Two Cards Drawn from $\{1, 3, 5, 7\}$)

- Let $X_1, X_2 \in \{1, 3, 5, 7\}$ be independent discrete uniform random variables representing the numbers on two cards drawn with replacement, each having the PMF

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{4}, & n \in \{1, 3, 5, 7\} \\ 0, & \text{otherwise} \end{cases} \tag{2.1.23.1}$$

- Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$,

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n - k \mid X_2 = k)\, p_{X_2}(k) \tag{2.1.23.2}$$

$$= \sum_k p_{X_1}(n - k)\, p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \tag{2.1.23.3}$$

- Substituting (2.1.23.1) into (2.1.23.3) yields

$$p_X(n) = \begin{cases} \frac{1}{16}, & n = 2, 14 \\ \frac{2}{16}, & n = 4, 12 \\ \frac{3}{16}, & n = 6, 10 \\ \frac{4}{16}, & n = 8 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.23.4}$$

- The cards are drawn without replacement, so the case $X_1 = X_2$ is not possible. It contributes $\frac{1}{16}$ to $n = 2, 6, 10, 14$, and $\Pr(X_1 = X_2) = \frac{1}{4}$. Removing it and

normalising by $1-\frac{1}{4}$ gives the PMF

$$p_X(n) = \begin{cases} \frac{1}{6}, & n = 4, 6, 10, 12 \\ \frac{1}{3}, & n = 8 \\ 0, & \text{otherwise} \end{cases} \tag{2.1.23.5}$$

The following code verifies (2.1.23.5) in Fig. 2.1.23.1

codes/PMF.py

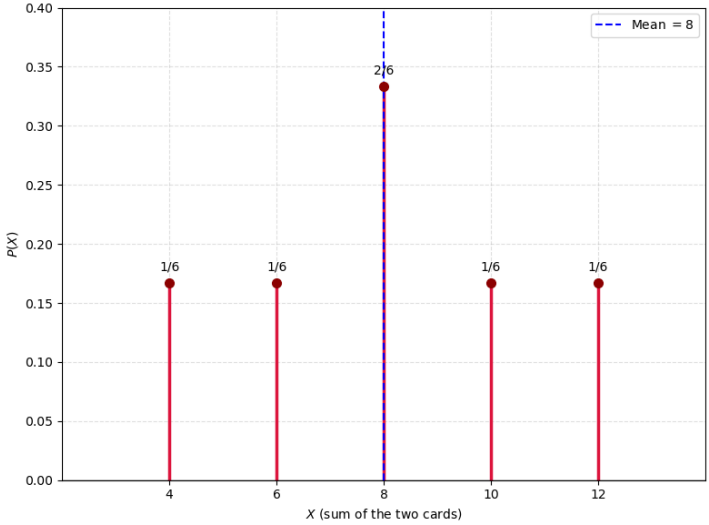


Fig. 2.1.23.1: Probability mass function

2.1.24 (Bayes' Theorem) Let $E_1, E_2, \ldots, E_k$ be mutually exclusive events covering the sample space. For an event A,

$$\Pr(E_i|A) = \frac{\Pr(E_i)\Pr(A|E_i)}{\sum_{j=1}^{k}\Pr(E_j)\Pr(A|E_j)} \tag{2.1.24.1}$$

2.1.25 The value of a determinant does not change if a multiple of one row (or column) is subtracted from another row (or column). A determinant can be expanded along any row or column, using the cofactors of its entries.

2.1.26 (System of Linear Equations) The matrix equation

$$\mathbf{A}\mathbf{x} = \mathbf{b} \tag{2.1.26.1}$$

is consistent if and only if

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \tag{2.1.26.2}$$

and the solution is unique if r equals the number of unknowns.

2.1.27 (Least Squares) If $\mathbf{Ax} = \mathbf{b}$ has more equations than unknowns, the least squares solution minimises

$$\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|^2 \implies \mathbf{x} = \left(\mathbf{A}^\top \mathbf{A}\right)^{-1} \mathbf{A}^\top \mathbf{b} \tag{2.1.27.1}$$

provided rank $(\mathbf{A})$ equals the number of unknowns. If the system is consistent, the minimum value is zero and $\mathbf{x}$ is the exact solution. **Solution:** Expanding the cost,

$$f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{Ax} - 2\mathbf{b}^\top \mathbf{Ax} + \mathbf{b}^\top \mathbf{b} \tag{2.1.27.2}$$

Setting the gradient to zero,

$$2\mathbf{A}^\top \mathbf{Ax} - 2\mathbf{A}^\top \mathbf{b} = \mathbf{0} \implies \mathbf{A}^\top \mathbf{Ax} = \mathbf{A}^\top \mathbf{b} \tag{2.1.27.3}$$

The matrix $\mathbf{A}^\top \mathbf{A}$ is invertible when rank $(\mathbf{A})$ equals the number of unknowns, which gives (2.1.27.1).

2.1.28 The matrix of the points $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{D}$ is $\mathbf{P} = \begin{pmatrix}\mathbf{A} & \mathbf{B} & \mathbf{C} & \mathbf{D}\end{pmatrix}$, and their direction matrix is

$$\mathbf{M} = \begin{pmatrix}\mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} & \mathbf{D}-\mathbf{A}\end{pmatrix} = \mathbf{P}\begin{pmatrix}-1 & -1 & -1\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix} \tag{2.1.28.1}$$

The points are coplanar if and only if

$$\text{rank}\,(\mathbf{M}) < 3 \tag{2.1.28.2}$$

that is, the determinant of $\mathbf{M}$ is zero.

2.1.29 The plane through the points $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ is

$$\mathbf{x} = \mathbf{A} + \lambda(\mathbf{B} - \mathbf{A}) + \mu(\mathbf{C} - \mathbf{A}), \quad \lambda, \mu \in \mathbb{R} \tag{2.1.29.1}$$

The point of intersection of this plane with the line in (2.1.30.1) is obtained by solving the linear system for λ, μ and κ, which has a unique solution if the rank of the coefficient matrix is 3.

2.1.30 The parametric form of the equation of a line is

$$\mathbf{x} = \mathbf{h} + \kappa\mathbf{m} \tag{2.1.30.1}$$

where $\mathbf{m}$ is the direction vector, and $\mathbf{m}^\top \mathbf{n} = 0$ for a normal $\mathbf{n}$ of the line.

2.1.31 (Line-Conic Intersection) The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ $(\kappa \in \mathbb{R})$ with a conic section $\mathbf{x}^\top \mathbf{Vx} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \tag{2.1.31.1}$$

where

$$\kappa_i = \frac{-\mathbf{m}^\top(\mathbf{Vh}+\mathbf{u}) \pm \sqrt{\left[\mathbf{m}^\top(\mathbf{Vh}+\mathbf{u})\right]^2 - g(\mathbf{h})(\mathbf{m}^\top \mathbf{Vm})}}{\mathbf{m}^\top \mathbf{Vm}}, \quad i = 1, 2 \tag{2.1.31.2}$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V}\mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

2.1.32 The area enclosed between the curves $y = f(x)$ (upper) and $y = g(x)$ (lower) from $x = a$ to $x = b$ is

$$A = \int_a^b [f(x) - g(x)]\, dx \tag{2.1.32.1}$$

2.1.33 A function $f : X \to Y$ is bijective if it is one-one and onto,

$$f(x_1) = f(x_2) \implies x_1 = x_2, \quad \forall y \in Y,\ \exists x \in X \text{ such that } f(x) = y \tag{2.1.33.1}$$

The inverse of a bijective function is obtained by solving $y = f(x)$ for x.

2.1.34 The function $f(x) = \frac{ax+b}{cx+d}$ corresponds to the matrix $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ acting on $\begin{pmatrix} x \\ 1 \end{pmatrix}$. Composition of such functions corresponds to the matrix product, and the inverse function corresponds to $\mathbf{M}^{-1}$ when $|\mathbf{M}| \neq 0$.

2.1.35 (Inverse by Gauss-Jordan Elimination) The inverse of an $n \times n$ matrix $\mathbf{A}$ exists if and only if $\text{rank}(\mathbf{A}) = n$. It is obtained by applying elementary row operations to the augmented matrix until the left half becomes the identity,

$$[\mathbf{A} \mid \mathbf{I}] \xrightarrow{\text{row operations}} [\mathbf{I} \mid \mathbf{A}^{-1}] \tag{2.1.35.1}$$

and the solution of $\mathbf{A}\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = \mathbf{A}^{-1}\mathbf{b}$. **Solution:** Each row operation is multiplication on the left by an elementary matrix. If $\mathbf{E}$ is the product of the elementary matrices that reduce $\mathbf{A}$ to $\mathbf{I}$, then

$$\mathbf{E}[\mathbf{A} \mid \mathbf{I}] = [\mathbf{E}\mathbf{A} \mid \mathbf{E}] = [\mathbf{I} \mid \mathbf{E}] \tag{2.1.35.2}$$

so $\mathbf{E}\mathbf{A} = \mathbf{I}$ and $\mathbf{E} = \mathbf{A}^{-1}$.

2.1.36 The Laplace transform of a function $g(t)$ is defined as

$$g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s) = \int_0^\infty e^{-st} g(t)\, dt \tag{2.1.36.1}$$

The independent variable can have any name, such as y.

2.1.37 The Laplace transform of the derivative is

$$g'(t) \overset{\mathcal{L}}{\longleftrightarrow} sG(s) - g(0) \tag{2.1.37.1}$$

Solution: Applying integration by parts, with $u = e^{-st}$ and $dv = g'(t)\, dt$,

$$\int_0^\infty e^{-st} g'(t)\, dt = \left[e^{-st} g(t)\right]_0^\infty + s \int_0^\infty e^{-st} g(t)\, dt = -g(0) + sG(s) \tag{2.1.37.2}$$

2.1.38 Multiplication by t corresponds to differentiation with respect to s,

$$t\, g(t) \overset{\mathcal{L}}{\longleftrightarrow} -\frac{dG(s)}{ds} \tag{2.1.38.1}$$

Solution: Differentiating (2.1.36.1) with respect to s,

$$\frac{dG(s)}{ds} = \int_0^\infty (-t)\, e^{-st} g(t)\, dt \tag{2.1.38.2}$$

2.1.39 For a non-negative integer n,

$$t^n u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{n!}{s^{n+1}} \tag{2.1.39.1}$$

Solution: From $u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s}$, applying (2.1.38.1) n times gives $(-1)^n \frac{d^n}{ds^n} \frac{1}{s} = \frac{n!}{s^{n+1}}$.

2.1.40 (Shift Property) For a constant a,

$$e^{at} g(t) \overset{\mathcal{L}}{\longleftrightarrow} G(s-a) \tag{2.1.40.1}$$

2.1.41 The Laplace transforms of the cosine and sine functions are

$$\cos(at)\, u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{s}{s^2+a^2}, \quad \sin(at)\, u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{a}{s^2+a^2} \tag{2.1.41.1}$$

2.1.42 The linear system $\frac{d\mathbf{x}}{dt} = \mathbf{M}\mathbf{x}$ with $\mathbf{x}(0) = \mathbf{x}_0$ has the Laplace transform

$$\mathbf{X}(s) = (s\mathbf{I} - \mathbf{M})^{-1} \mathbf{x}_0 \tag{2.1.42.1}$$

Solution: Taking the Laplace transform of each row and using (2.1.37.1),

$$s\mathbf{X}(s) - \mathbf{x}_0 = \mathbf{M}\mathbf{X}(s) \implies (s\mathbf{I} - \mathbf{M})\mathbf{X}(s) = \mathbf{x}_0 \tag{2.1.42.2}$$

2.1.43 For a 2×2 matrix $\mathbf{M}$, the poles of $(s\mathbf{I} - \mathbf{M})^{-1}$ are the eigenvalues of $\mathbf{M}$, and

$$\left|s\mathbf{I} - \mathbf{M}\right| = s^2 - \operatorname{tr}(\mathbf{M})\, s + \left|\mathbf{M}\right| \tag{2.1.43.1}$$

The sum of the eigenvalues is $\operatorname{tr}(\mathbf{M})$ and their product is $\left|\mathbf{M}\right|$.

B. Problems

2.2.1 Find the distance between the planes $2x - y + 2z = 5$ and $5x - 2.5y + 5z = 20$. (12, 2017)

Solution: The first plane is, from (2.1.1.1),

$$\begin{pmatrix} 2 & -1 & 2 \end{pmatrix} \mathbf{x} = 5 \tag{2.2.1.1}$$

Dividing the second equation by 2.5,

$$\begin{pmatrix} 5 & -2.5 & 5 \end{pmatrix} \mathbf{x} = 20 \implies \begin{pmatrix} 2 & -1 & 2 \end{pmatrix} \mathbf{x} = 8 \tag{2.2.1.2}$$

The planes are parallel with

$$\mathbf{n} = \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix}, \quad c_1 = 5, \quad c_2 = 8, \quad \|\mathbf{n}\| = \sqrt{4+1+4} = 3 \tag{2.2.1.3}$$

From (2.1.2.1),

$$d = \frac{|8-5|}{3} = 1 \tag{2.2.1.4}$$

The following code verifies the solution

codes/distance.py

2.2.2 If for any 2×2 square matrix $\mathbf{A}$, $\mathbf{A}\,\mathrm{adj}\,(\mathbf{A}) = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix}$, then write the value of $\left|\mathbf{A}\right|$. (12, 2017)

Solution: From (2.1.3.1),

$$\mathbf{A}\,\mathrm{adj}\,(\mathbf{A}) = \left|\mathbf{A}\right|\mathbf{I} = \begin{pmatrix} 8 & 0 \\ 0 & 8 \end{pmatrix} = 8\mathbf{I} \tag{2.2.2.1}$$

Comparing the entries,

$$\left|\mathbf{A}\right| = 8 \tag{2.2.2.2}$$

As a check, taking the trace of both sides using (2.1.4.1),

$$\mathrm{tr}\left(\left|\mathbf{A}\right|\mathbf{I}\right) = 2\left|\mathbf{A}\right| = \mathrm{tr}\,(8\mathbf{I}) = 16 \implies \left|\mathbf{A}\right| = 8 \tag{2.2.2.3}$$

The following code verifies the solution

codes/determinant.py

2.2.3 A die, whose faces are marked 1, 2, 3 in red and 4, 5, 6 in green, is tossed. Let A be the event "number obtained is even" and B be the event "number obtained is red". Find if A and B are independent events. (12, 2017)

Solution: The events are

$$A = \{2, 4, 6\}, \quad B = \{1, 2, 3\}, \quad AB = \{2\} \tag{2.2.3.1}$$

From (2.1.5.1) and (2.1.5.2),

$$\Pr(A) = \frac{3}{6} = \frac{1}{2}, \quad \Pr(B) = \frac{3}{6} = \frac{1}{2}, \quad \Pr(AB) = \frac{1}{6} \tag{2.2.3.2}$$

From (2.1.6.1),

$$\Pr(A)\Pr(B) = \frac{1}{4} \neq \frac{1}{6} = \Pr(AB) \tag{2.2.3.3}$$

Therefore, A and B are not independent events.
The following code verifies the solution

codes/dice.c

2.2.4 Two tailors, A and B, earn ₹ 300 and ₹ 400 per day respectively. A can stitch 6 shirts and 4 pairs of trousers while B can stitch 10 shirts and 4 pairs of trousers per day. To find how many days should each of them work and if it is desired to produce at least 60 shirts and 32 pairs of trousers at a minimum labour cost, formulate this as an LPP. (12, 2017)

Solution: The following code generates Fig. 2.2.4.1

codes/tailor_cost.py

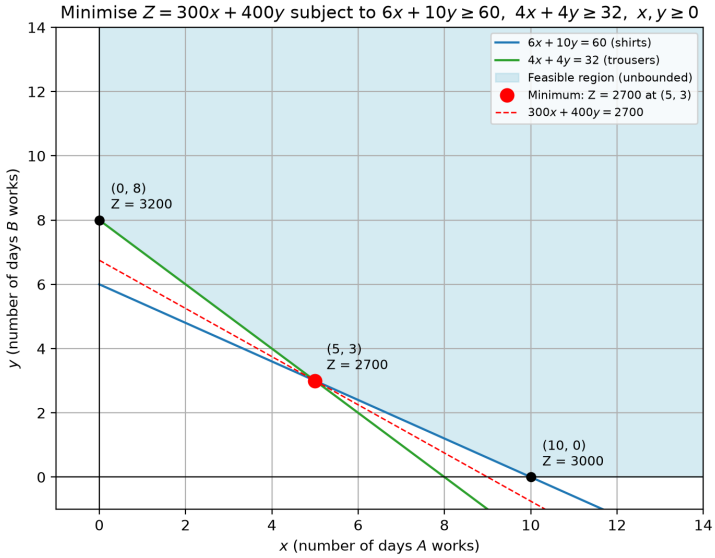


Fig. 2.2.4.1: Feasible region of the LPP

2.2.5 Find the value of c in Rolle's theorem for the function $f(x) = x^3 - 3x$ in $\left[-\sqrt{3}, 0\right]$. (12, 2017)

Solution: The function $f(x)$ is a polynomial, so it is differentiable, and hence continuous, from (2.1.9.1). At the end points,

$$f\left(-\sqrt{3}\right) = -3\sqrt{3} + 3\sqrt{3} = 0, \quad f(0) = 0 \tag{2.2.5.1}$$

The conditions of (2.1.13.1) hold, so there exists $c \in \left(-\sqrt{3}, 0\right)$ with $f'(c) = 0$. The derivative is

$$f'(x) = 3x^2 - 3 = 0 \tag{2.2.5.2}$$

which is a quadratic equation. To solve it, apply (2.1.11.1) to $g(x) = f'(x) = 3x^2 - 3$, which is of the form $ax^2 + bx + c$ with $a = 3$, $b = 0$, $c = -3$. From (2.1.12.1), the difference equation is

$$x_{n+1} = \frac{3x_n^2 + 3}{6x_n} = \frac{x_n^2 + 1}{2x_n} \tag{2.2.5.3}$$

Starting with $x_0 = -1.5 \in \left(-\sqrt{3}, 0\right)$,

$$x_1 \approx -1.0833, \quad x_2 \approx -1.0032, \quad x_3 \approx -1.0000 \tag{2.2.5.4}$$

The iterations converge to -1. This can be justified from the difference equation, since

$$x_{n+1}+1=\frac{(x_n+1)^2}{2x_n} \tag{2.2.5.5}$$

For $x_n<0$, the right side is non-positive, so $x_{n+1}\le -1$ after the first step. Also, $|x_n+1|<2|x_n|$, so $|x_{n+1}+1|<|x_n+1|$ and the error keeps decreasing. Hence $x_n\to -1$, which is the same as the root of $3(x-1)(x+1)=0$ lying in $\left(-\sqrt{3},0\right)$, since $-\sqrt{3}\approx -1.732$ and $+1$ is outside the interval. Therefore,

$$c=-1 \tag{2.2.5.6}$$

The tangent at $(c,f(c))=(-1,2)$ is horizontal, and it is parallel to the chord joining the end points, as shown in Fig. 2.2.5.1.
The following code verifies the solution using the difference equation

codes/rolle.c

The following code generates Fig. 2.2.5.1

codes/rolle.py

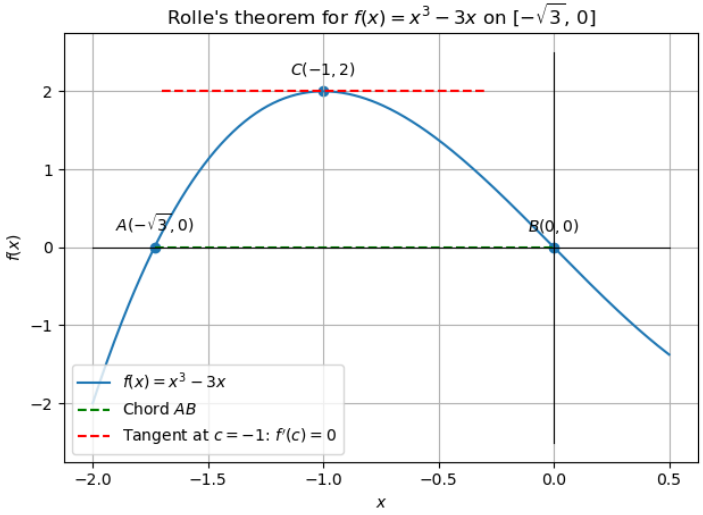


Fig. 2.2.5.1: Graph of $f(x)=x^3-3x$ on $\left[-\sqrt{3},0\right]$ with the horizontal tangent at $c=-1$

2.2.6 Show that the function $f(x)=x^3-3x^2+6x-100$ is increasing on $\mathbb{R}$. (12, 2017)

Solution: Differentiating,

$$f'(x) = 3x^2 - 6x + 6 = 3\left[(x-1)^2 + 1\right] \tag{2.2.6.1}$$

Since $(x-1)^2 \geq 0$ for all $x \in \mathbb{R}$,

$$f'(x) \geq 3 > 0 \tag{2.2.6.2}$$

Hence, from (2.1.14), $f(x)$ is strictly increasing on $\mathbb{R}$.
The following code generates Fig. 2.2.6.1

codes/Increasing.py

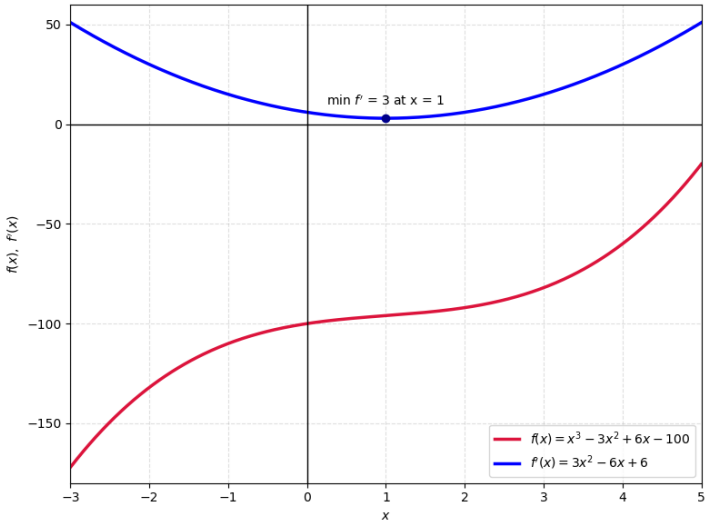


Fig. 2.2.6.1: Increasing function

2.2.7 Evaluate :

$$\int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx$$

(12, 2017)

Solution: Let

$$I = \int_0^{\pi} \frac{x \tan x}{\sec x + \tan x}\, dx \tag{2.2.7.1}$$

From (2.1.16.1) and (2.1.17.1),

$$I = \int_0^{\pi} \frac{(\pi - x)(-\tan x)}{-\sec x - \tan x}\, dx = \int_0^{\pi} \frac{(\pi - x)\tan x}{\sec x + \tan x}\, dx \tag{2.2.7.2}$$

Adding the two expressions of I,

$$2I = \pi \int_0^\pi \frac{\tan x}{\sec x + \tan x}\, dx = \pi \int_0^\pi \frac{\sin x}{1+\sin x}\, dx = \pi \int_0^\pi \left[1 - \frac{1}{1+\sin x}\right] dx \tag{2.2.7.3}$$

from (2.1.17.2). Using (2.1.17.2) and (2.1.18.1),

$$\int_0^\pi \frac{dx}{1+\sin x} = \int_0^\pi \left(\sec^2 x - \sec x \tan x\right) dx = \left[\tan x - \sec x\right]_0^\pi \tag{2.2.7.4}$$

Here $\tan x - \sec x = -\frac{\cos x}{1+\sin x}$ is continuous on $[0, \pi]$, so

$$\left[-\frac{\cos x}{1+\sin x}\right]_0^\pi = 1 - (-1) = 2 \tag{2.2.7.5}$$

Therefore,

$$2I = \pi\,(\pi - 2) \implies I = \frac{\pi\,(\pi-2)}{2} \approx 1.7932 \tag{2.2.7.6}$$

To verify this numerically, from (2.1.17.2) the integrand is

$$f\,(x) = \frac{x \tan x}{\sec x + \tan x} = \frac{x \sin x}{1+\sin x} \tag{2.2.7.7}$$

which is continuous on $[0, \pi]$, even though $\tan x$ is not defined at $x = \frac{\pi}{2}$. Taking N equal sub-intervals of width $h = \frac{\pi}{N}$, with $x_n = nh$, the difference equation of (2.1.15.1) is

$$I_{n+1} = I_n + \frac{h}{2}\left(f\,(x_n) + f\,(x_{n+1})\right), \quad I_0 = 0 \tag{2.2.7.8}$$

and $I \approx I_N$. As N increases, I_N approaches $\frac{\pi(\pi-2)}{2}$, with an error that decreases as h^2.

The following code verifies the solution using the difference equation

codes/Integral_xtanx.c

The following code generates Fig. 2.2.7.1

codes/Integral_xtanx.py

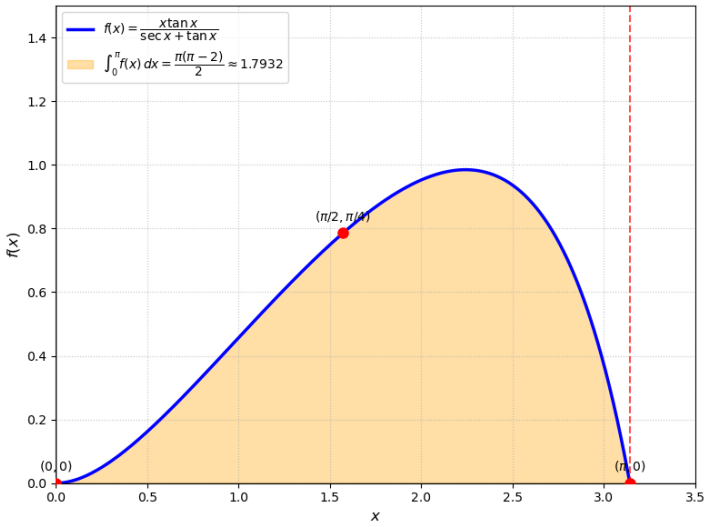


Fig. 2.2.7.1: Integral of $x \tan x$

2.2.8 Evaluate :

$$\int_1^4 \{|x-1| + |x-2| + |x-4|\}\, dx$$

(12, 2017)

Solution: From (2.1.19.1), for $1 \le x \le 4$, $|x-1| = x-1$ and $|x-4| = 4-x$, while $|x-2|$ changes at $x = 2$. Thus the integrand is

$$g(x) = \begin{cases} (x-1) + (2-x) + (4-x) = 5-x, & 1 \le x \le 2 \\ (x-1) + (x-2) + (4-x) = x+1, & 2 < x \le 4 \end{cases} \tag{2.2.8.1}$$

At the boundary $x = 2$, from (2.1.10.2),

$$f_1(2) = 5 - 2 = 3 = 2 + 1 = f_2(2) \tag{2.2.8.2}$$

so $g(x)$ is continuous. From (2.1.10.3), $f_1'(2) = -1 \neq 1 = f_2'(2)$, so $g(x)$ is not differentiable at $x = 2$, but it is continuous and hence integrable. Splitting the integral at $x = 2$,

$$\int_1^2 (5-x)\, dx = \left[5x - \frac{x^2}{2}\right]_1^2 = 8 - \frac{9}{2} = \frac{7}{2} \tag{2.2.8.3}$$

$$\int_2^4 (x+1)\, dx = \left[\frac{x^2}{2} + x\right]_2^4 = 12 - 4 = 8 \tag{2.2.8.4}$$

Adding,

$$\int_1^4 \{|x-1|+|x-2|+|x-4|\}\,dx = \frac{7}{2}+8 = \frac{23}{2} \tag{2.2.8.5}$$

The following code generates Fig. 2.2.8.1

codes/Integral_modulus.py

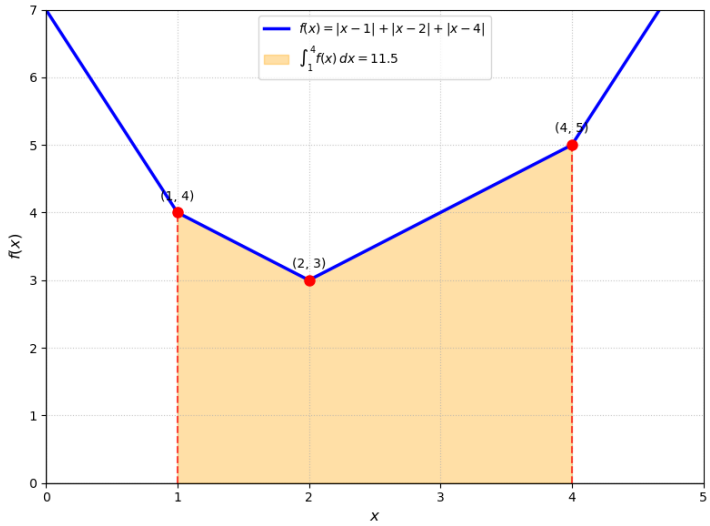


Fig. 2.2.8.1: Integral of the modulus function

2.2.9 Show that the points A, B, C with position vectors $2\hat{i}-\hat{j}+\hat{k}$, $\hat{i}-3\hat{j}-5\hat{k}$ and $3\hat{i}-4\hat{j}-4\hat{k}$ respectively, are the vertices of a right-angled triangle. Hence find the area of the triangle. (12, 2017)

Solution: The position vectors are

$$\mathbf{A} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 1 \\ -3 \\ -5 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 3 \\ -4 \\ -4 \end{pmatrix} \tag{2.2.9.1}$$

The sides are

$$\mathbf{B}-\mathbf{A} = \begin{pmatrix} -1 \\ -2 \\ -6 \end{pmatrix}, \quad \mathbf{C}-\mathbf{B} = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}, \quad \mathbf{A}-\mathbf{C} = \begin{pmatrix} -1 \\ 3 \\ 5 \end{pmatrix} \tag{2.2.9.2}$$

From (2.1.20.1),

$$(\mathbf{B}-\mathbf{A})^{\top}(\mathbf{C}-\mathbf{B}) = -2+2-6 = -6 \neq 0 \tag{2.2.9.3}$$

$$(\mathbf{C}-\mathbf{B})^{\top}(\mathbf{A}-\mathbf{C}) = -2-3+5 = 0 \tag{2.2.9.4}$$

$$(\mathbf{A}-\mathbf{C})^{\top}(\mathbf{B}-\mathbf{A}) = 1-6-30 = -35 \neq 0 \tag{2.2.9.5}$$

Hence $CB \perp CA$, and the triangle is right angled at C. Also,

$$\|\mathbf{C}-\mathbf{B}\|^2 + \|\mathbf{A}-\mathbf{C}\|^2 = 6+35 = 41 = \|\mathbf{B}-\mathbf{A}\|^2 \tag{2.2.9.6}$$

From (2.1.21.1),

$$(\mathbf{C}-\mathbf{B}) \times (\mathbf{A}-\mathbf{C}) = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \times \begin{pmatrix} -1 \\ 3 \\ 5 \end{pmatrix} = \begin{pmatrix} -8 \\ -11 \\ 5 \end{pmatrix} \tag{2.2.9.7}$$

so the area is

$$A = \frac{1}{2}\left\| \begin{pmatrix} -8 \\ -11 \\ 5 \end{pmatrix} \right\| = \frac{1}{2}\sqrt{64+121+25} = \frac{\sqrt{210}}{2} \text{ sq. units} \tag{2.2.9.8}$$

The following code verifies the solution

codes/triangle.py

2.2.10 There are 4 cards numbered 1, 3, 5 and 7, one number on one card. Two cards are drawn at random without replacement. Let X denote the sum of the numbers on the two drawn cards. Find the mean and variance of X. (12, 2017)

Solution: Let X_1 and X_2 represent the numbers on the two drawn cards, and let $X = X_1 + X_2$ denote their sum. From the PMF derived in (2.1.23.5),

$$p_X(4) = p_X(6) = p_X(10) = p_X(12) = \frac{1}{6}, \quad p_X(8) = \frac{1}{3} \tag{2.2.10.1}$$

From (2.1.22.1), the mean is

$$E(X) = \frac{4+6+2(8)+10+12}{6} = 8 \tag{2.2.10.2}$$

and

$$\sum_n n^2 p_X(n) = \frac{16+36+2(64)+100+144}{6} = \frac{212}{3} \tag{2.2.10.3}$$

Therefore, the variance is

$$\text{Var}(X) = \frac{212}{3} - 64 = \frac{20}{3} \tag{2.2.10.4}$$

2.2.11 Of the students in a school, it is known that 30% have 100% attendance and 70% students are irregular. Previous year results report that 70% of all students who have 100% attendance attain A grade and 10% irregular students attain A grade in their annual examination. At the end of the year, one student is chosen at random from

the school and he was found to have an A grade. What is the probability that the student has 100% attendance ? Is regularity required only in school ? Justify your answer. (12, 2017)

Solution: Let E_1 be the event that the student has 100% attendance, E_2 be the event that the student is irregular and A be the event that the student has an A grade. Then

$$\Pr(E_1) = \frac{3}{10}, \quad \Pr(E_2) = \frac{7}{10}, \quad \Pr(A|E_1) = \frac{7}{10}, \quad \Pr(A|E_2) = \frac{1}{10} \tag{2.2.11.1}$$

From (2.1.24.1),

$$\Pr(E_1|A) = \frac{\frac{3}{10}\cdot\frac{7}{10}}{\frac{3}{10}\cdot\frac{7}{10}+\frac{7}{10}\cdot\frac{1}{10}} = \frac{21}{21+7} = \frac{3}{4} \tag{2.2.11.2}$$

Regularity is not required only in school. It is needed in every area of life, such as studies, work, sports and health habits, because steady and regular effort gives better results. This is seen in the data above, where 70% of the regular students get an A grade against 10% of the irregular students. The following code verifies the solution

codes/bayes.c

2.2.12 Find matrix $\mathbf{A}$ such that

$$\begin{pmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{pmatrix} \mathbf{A} = \begin{pmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{pmatrix}$$

(12, 2017)

Solution: Let the equation be $\mathbf{BA} = \mathbf{C}$, where $\mathbf{B}$ is 3×2 and $\mathbf{A}$ is 2×2. The columns of $\mathbf{A}$ satisfy $\mathbf{Ba}_j = \mathbf{c}_j$ for $j = 1, 2$, which is an overdetermined system. From (2.1.27.1),

$$\mathbf{A} = \left(\mathbf{B}^\top\mathbf{B}\right)^{-1}\mathbf{B}^\top\mathbf{C} \tag{2.2.12.1}$$

where

$$\mathbf{B}^\top\mathbf{B} = \begin{pmatrix} 2 & 1 & -3 \\ -1 & 0 & 4 \end{pmatrix}\begin{pmatrix} 2 & -1 \\ 1 & 0 \\ -3 & 4 \end{pmatrix} = \begin{pmatrix} 14 & -14 \\ -14 & 17 \end{pmatrix} \tag{2.2.12.2}$$

with $\left|\mathbf{B}^\top\mathbf{B}\right| = 238 - 196 = 42 \neq 0$, so

$$\left(\mathbf{B}^\top\mathbf{B}\right)^{-1} = \frac{1}{42}\begin{pmatrix} 17 & 14 \\ 14 & 14 \end{pmatrix} \tag{2.2.12.3}$$

Also,

$$\mathbf{B}^\top\mathbf{C} = \begin{pmatrix} 2 & 1 & -3 \\ -1 & 0 & 4 \end{pmatrix}\begin{pmatrix} -1 & -8 \\ 1 & -2 \\ 9 & 22 \end{pmatrix} = \begin{pmatrix} -28 & -84 \\ 37 & 96 \end{pmatrix} \tag{2.2.12.4}$$

Therefore,

$$\mathbf{A} = \frac{1}{42}\begin{pmatrix}17 & 14\\ 14 & 14\end{pmatrix}\begin{pmatrix}-28 & -84\\ 37 & 96\end{pmatrix} = \frac{1}{42}\begin{pmatrix}42 & -84\\ 126 & 168\end{pmatrix} = \begin{pmatrix}1 & -2\\ 3 & 4\end{pmatrix} \quad (2.2.12.5)$$

As a check, $\mathbf{BA} = \mathbf{C}$, so the residual $\|\mathbf{B}\mathbf{a}_j - \mathbf{c}_j\|$ is zero for both columns and the least squares solution is the exact solution.
The following code verifies the solution

codes/matrix.py

2.2.13 Solve the following linear programming problem graphically :
Maximise $Z = 34x + 45y$ under the following constraints

$$x + y \leq 300, \quad 2x + 3y \leq 70, \quad x \geq 0,\ y \geq 0$$

(12, 2017)

Solution: The following code generates Fig. 2.2.13.1

codes/LPP.py

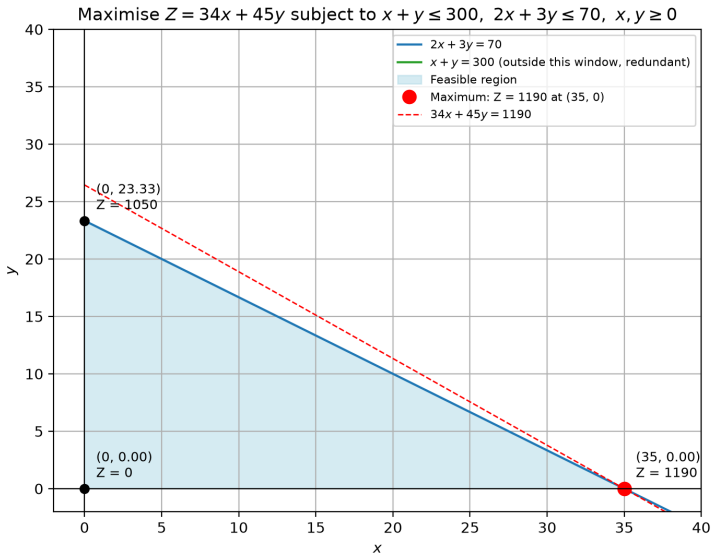


Fig. 2.2.13.1: Linear programming problem

2.2.14 Find the value of x such that the points $A\,(3, 2, 1)$, $B\,(4, x, 5)$, $C\,(4, 2, -2)$ and $D\,(6, 5, -1)$ are coplanar. (12, 2017)

Solution: The matrix of the points is

$$\mathbf{P} = \begin{pmatrix} \mathbf{A} & \mathbf{B} & \mathbf{C} & \mathbf{D} \end{pmatrix} = \begin{pmatrix} 3 & 4 & 4 & 6 \\ 2 & x & 2 & 5 \\ 1 & 5 & -2 & -1 \end{pmatrix} \tag{2.2.14.1}$$

From (2.1.28.1), the direction matrix is

$$\mathbf{M} = \begin{pmatrix} \mathbf{B}-\mathbf{A} & \mathbf{C}-\mathbf{A} & \mathbf{D}-\mathbf{A} \end{pmatrix} = \mathbf{P}\begin{pmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 3 \\ x-2 & 0 & 3 \\ 4 & -3 & -2 \end{pmatrix} \tag{2.2.14.2}$$

The points are coplanar if rank $(\mathbf{M}) < 3$, that is, $|\mathbf{M}| = 0$. Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - 3C_1$, from (2.1.25),

$$|\mathbf{M}| = \begin{vmatrix} 1 & 0 & 0 \\ x-2 & -(x-2) & 9-3x \\ 4 & -7 & -14 \end{vmatrix} = 14\,(x-2) + 7\,(9-3x) = 35 - 7x \tag{2.2.14.3}$$

Equating to zero,

$$35 - 7x = 0 \implies x = 5 \tag{2.2.14.4}$$

As a check, for $x = 5$ the columns of $\mathbf{M}$ satisfy $\begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} + 2\begin{pmatrix} 1 \\ 0 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \\ -2 \end{pmatrix}$, so the rank is less than 3.
The following code verifies the solution

codes/coplanar.py

2.2.15 Find the general solution of the differential equation

$$y\,dx - \left(x + 2y^2\right)\,dy = 0$$

(12, 2017)

Solution: Rewriting the equation and multiplying by $y > 0$,

$$\frac{dx}{dy} - \frac{x}{y} = 2y \implies y\frac{dx}{dy} - x = 2y^2 \tag{2.2.15.1}$$

Let $x\,(y) \overset{\mathcal{L}}{\longleftrightarrow} X\,(s)$. From (2.1.38.1) and (2.1.37.1),

$$y\frac{dx}{dy} \overset{\mathcal{L}}{\longleftrightarrow} -\frac{d}{ds}\left[sX\,(s) - x\,(0)\right] = -X\,(s) - sX'\,(s) \tag{2.2.15.2}$$

From (2.1.39.1), $2y^2 \overset{\mathcal{L}}{\longleftrightarrow} \frac{4}{s^3}$. Taking the Laplace transform of (2.2.15.1),

$$-X - sX' - X = \frac{4}{s^3} \tag{2.2.15.3}$$

$$\implies X' + \frac{2}{s}X = -\frac{4}{s^4} \tag{2.2.15.4}$$

Multiplying by s^2,

$$\frac{d}{ds}\left(s^2 X\right) = -\frac{4}{s^2} \tag{2.2.15.5}$$

$$\implies s^2 X = \frac{4}{s} + C \tag{2.2.15.6}$$

$$\implies X(s) = \frac{4}{s^3} + \frac{C}{s^2} \tag{2.2.15.7}$$

Taking the inverse Laplace transform using (2.1.39.1),

$$x(y) = 2y^2 + Cy \tag{2.2.15.8}$$

Note that $x(0)$ cancelled out, so C remains arbitrary. As a check, $\frac{dx}{dy} = 4y + C$ and $\frac{x}{y} + 2y = 2y + C + 2y = 4y + C$. Therefore, the general solution is

$$x = 2y^2 + Cy \tag{2.2.15.9}$$

2.2.16 Find the coordinates of the point where the line through the points $(3, -4, -5)$ and $(2, -3, 1)$, crosses the plane determined by the points $(1, 2, 3)$, $(4, 2, -3)$ and $(0, 4, 3)$. (12, 2017)

Solution: Let $\mathbf{A}$, $\mathbf{B}$, $\mathbf{C}$ be the points of the plane and $\mathbf{D}$, $\mathbf{E}$ be the points of the line. The matrix of the points is

$$\mathbf{P} = \begin{pmatrix}\mathbf{A} & \mathbf{B} & \mathbf{C} & \mathbf{D} & \mathbf{E}\end{pmatrix} = \begin{pmatrix} 1 & 4 & 0 & 3 & 2 \\ 2 & 2 & 4 & -4 & -3 \\ 3 & -3 & 3 & -5 & 1 \end{pmatrix} \tag{2.2.16.1}$$

From (2.1.29.1) and (2.1.30.1), the point of intersection satisfies

$$\mathbf{A} + \lambda(\mathbf{B} - \mathbf{A}) + \mu(\mathbf{C} - \mathbf{A}) = \mathbf{D} + \kappa(\mathbf{E} - \mathbf{D}) \tag{2.2.16.2}$$

which can be written as

$$\begin{pmatrix}\mathbf{B} - \mathbf{A} & \mathbf{C} - \mathbf{A} & \mathbf{D} - \mathbf{E}\end{pmatrix} \begin{pmatrix}\lambda \\ \mu \\ \kappa\end{pmatrix} = \mathbf{D} - \mathbf{A} \tag{2.2.16.3}$$

The direction matrix

$$\mathbf{M} = \begin{pmatrix}\mathbf{B} - \mathbf{A} & \mathbf{C} - \mathbf{A} & \mathbf{D} - \mathbf{E} & \mathbf{D} - \mathbf{A}\end{pmatrix} \tag{2.2.16.4}$$

is obtained from $\mathbf{P}$ as

$$\mathbf{M} = \mathbf{P}\begin{pmatrix} -1 & -1 & 0 & -1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & -1 & 0 \end{pmatrix} = \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ -6 & 0 & -6 & -8 \end{pmatrix} \tag{2.2.16.5}$$

Row reducing the augmented matrix $\mathbf{M}$,

$$\begin{aligned} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ -6 & 0 & -6 & -8 \end{pmatrix} &\xrightarrow{R_3 \to R_3 + 2R_1} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ 0 & -2 & -4 & -4 \end{pmatrix} \\ &\xrightarrow{R_3 \to R_3 + R_2} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ 0 & 0 & -5 & -10 \end{pmatrix} \\ &\xrightarrow{R_3 \to -\frac{1}{5}R_3} \begin{pmatrix} 3 & -1 & 1 & 2 \\ 0 & 2 & -1 & -6 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow[R_1 \to R_1 - R_3]{R_2 \to R_2 + R_3} \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 2 & 0 & -4 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow{R_2 \to \frac{1}{2}R_2} \begin{pmatrix} 3 & -1 & 0 & 0 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow{R_1 \to R_1 + R_2} \begin{pmatrix} 3 & 0 & 0 & -2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \\ &\xrightarrow{R_1 \to \frac{1}{3}R_1} \begin{pmatrix} 1 & 0 & 0 & -\frac{2}{3} \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 2 \end{pmatrix} \end{aligned} \tag{2.2.16.6}$$

Here the rank of the coefficient matrix and the augmented matrix is 3, which is the number of unknowns. From (2.1.26.2), the solution is unique,

$$\lambda = -\frac{2}{3}, \quad \mu = -2, \quad \kappa = 2 \tag{2.2.16.7}$$

Therefore, the point of intersection is

$$\mathbf{x} = \mathbf{D} + \kappa\left(\mathbf{E} - \mathbf{D}\right) = \begin{pmatrix} 3 \\ -4 \\ -5 \end{pmatrix} + 2\begin{pmatrix} -1 \\ 1 \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \\ 7 \end{pmatrix} \tag{2.2.16.8}$$

As a check,

$$\mathbf{A}+\lambda\left(\mathbf{B}-\mathbf{A}\right)+\mu\left(\mathbf{C}-\mathbf{A}\right)=\begin{pmatrix}1\\2\\3\end{pmatrix}-\frac{2}{3}\begin{pmatrix}3\\0\\-6\end{pmatrix}-2\begin{pmatrix}-1\\2\\0\end{pmatrix}=\begin{pmatrix}1\\-2\\7\end{pmatrix} \tag{2.2.16.9}$$

The following code verifies the solution

codes/lineplane.py

2.2.17 Find the particular solution of the differential equation $(x-y)\frac{dy}{dx}=(x+2y)$, given that $y=0$ when $x=1$. (12, 2017)

Solution: The equation contains the product $y\frac{dy}{dx}$, so it cannot be transformed directly. Introducing a parameter t such that

$$\frac{dx}{dt}=x-y,\quad \frac{dy}{dt}=x+2y \tag{2.2.17.1}$$

gives $\frac{dy}{dx}=\frac{dy/dt}{dx/dt}=\frac{x+2y}{x-y}$, which is the given equation. In matrix form,

$$\frac{d}{dt}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}1&-1\\1&2\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix},\quad \begin{pmatrix}x\\y\end{pmatrix}_{t=0}=\begin{pmatrix}1\\0\end{pmatrix} \tag{2.2.17.2}$$

Let $\mathbf{M}=\begin{pmatrix}1&-1\\1&2\end{pmatrix}$. Then $\text{tr}\left(\mathbf{M}\right)=3$ and $\left|\mathbf{M}\right|=3$. From (2.1.43.1),

$$\left|s\mathbf{I}-\mathbf{M}\right|=s^2-3s+3=\left(s-\frac{3}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2 \tag{2.2.17.3}$$

so the eigenvalues are $\frac{3\pm j\sqrt{3}}{2}$. From (2.1.42.1),

$$\begin{pmatrix}X\left(s\right)\\Y\left(s\right)\end{pmatrix}=\begin{pmatrix}s-1&1\\-1&s-2\end{pmatrix}^{-1}\begin{pmatrix}1\\0\end{pmatrix}=\frac{1}{s^2-3s+3}\begin{pmatrix}s-2\\1\end{pmatrix} \tag{2.2.17.4}$$

Writing $s-2=\left(s-\frac{3}{2}\right)-\frac{1}{2}$,

$$X\left(s\right)=\frac{s-\frac{3}{2}}{\left(s-\frac{3}{2}\right)^2+\frac{3}{4}}-\frac{1}{\sqrt{3}}\cdot\frac{\frac{\sqrt{3}}{2}}{\left(s-\frac{3}{2}\right)^2+\frac{3}{4}} \tag{2.2.17.5}$$

$$Y\left(s\right)=\frac{2}{\sqrt{3}}\cdot\frac{\frac{\sqrt{3}}{2}}{\left(s-\frac{3}{2}\right)^2+\frac{3}{4}} \tag{2.2.17.6}$$

Taking the inverse Laplace transform using (2.1.40.1) and (2.1.41.1), with $\theta=\frac{\sqrt{3}}{2}t$,

$$x\left(t\right)=e^{\frac{3t}{2}}\left(\cos\theta-\frac{1}{\sqrt{3}}\sin\theta\right) \tag{2.2.17.7}$$

$$y\left(t\right)=\frac{2}{\sqrt{3}}e^{\frac{3t}{2}}\sin\theta \tag{2.2.17.8}$$

To eliminate t,

$$x+2y=2e^{\frac{3t}{2}}\sin\left(\theta+\frac{\pi}{6}\right),\quad \sqrt{3}x=2e^{\frac{3t}{2}}\cos\left(\theta+\frac{\pi}{6}\right) \tag{2.2.17.9}$$

$$\implies \frac{x+2y}{\sqrt{3}x}=\tan\left(\theta+\frac{\pi}{6}\right) \tag{2.2.17.10}$$

$$\implies \tan^{-1}\left(\frac{x+2y}{\sqrt{3}x}\right)=\frac{\sqrt{3}}{2}t+\frac{\pi}{6} \tag{2.2.17.11}$$

Also, since $x+\frac{y}{2}=e^{\frac{3t}{2}}\cos\theta$,

$$x^2+xy+y^2=\left(x+\frac{y}{2}\right)^2+\frac{3}{4}y^2=e^{3t}\left(\cos^2\theta+\sin^2\theta\right)=e^{3t} \tag{2.2.17.12}$$

$$\implies \frac{1}{2}\ln\left(x^2+xy+y^2\right)=\frac{3t}{2} \tag{2.2.17.13}$$

Eliminating t from the last two results, the particular solution is

$$\sqrt{3}\tan^{-1}\left(\frac{x+2y}{\sqrt{3}x}\right)-\frac{1}{2}\ln\left(x^2+xy+y^2\right)=\frac{\pi}{2\sqrt{3}} \tag{2.2.17.14}$$

which holds on the branch through $(1,0)$. As a check, at $(1,0)$ the left side is $\sqrt{3}\cdot\frac{\pi}{6}-0=\frac{\pi}{2\sqrt{3}}$.
The following code generates Fig. 2.2.17.1

codes/euler.py

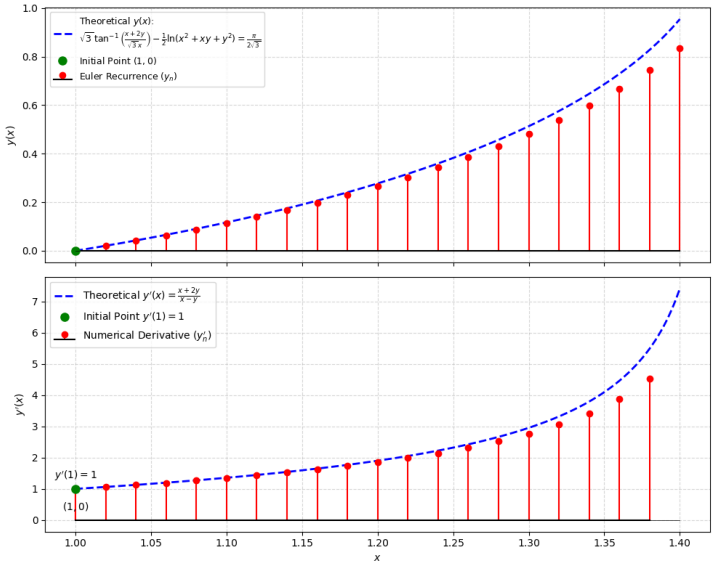


Fig. 2.2.17.1: Euler's method

2.2.18 Find the area enclosed between the parabola $4y = 3x^2$ and the straight line $3x - 2y + 12 = 0$. (12, 2017)

Solution: The parabola $3x^2 - 4y = 0$ is a conic with

$$\mathbf{V} = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad f = 0 \tag{2.2.18.1}$$

The line $3x - 2y + 12 = 0$ is $\mathbf{x} = \mathbf{h} + \kappa\mathbf{m}$ with

$$\mathbf{h} = \begin{pmatrix} 0 \\ 6 \end{pmatrix}, \quad \mathbf{m} = \begin{pmatrix} 2 \\ 3 \end{pmatrix} \tag{2.2.18.2}$$

Then

$$\mathbf{Vh} + \mathbf{u} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}, \quad \mathbf{m}^\top (\mathbf{Vh} + \mathbf{u}) = -6, \quad g(\mathbf{h}) = -24, \quad \mathbf{m}^\top \mathbf{Vm} = 12 \tag{2.2.18.3}$$

From (2.1.31.1),

$$\kappa = \frac{6 \pm \sqrt{36 + 288}}{12} = \frac{6 \pm 18}{12} = 2 \text{ or } -1 \tag{2.2.18.4}$$

The points of intersection are

$$\begin{pmatrix} 0 \\ 6 \end{pmatrix} + 2\begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 12 \end{pmatrix}, \quad \begin{pmatrix} 0 \\ 6 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ 3 \end{pmatrix} \tag{2.2.18.5}$$

The line is above the parabola between $x = -2$ and $x = 4$. From (2.1.32.1),

$$A = \int_{-2}^{4} \left[\frac{3x+12}{2} - \frac{3x^2}{4}\right] dx = \left[\frac{3x^2}{4} + 6x - \frac{x^3}{4}\right]_{-2}^{4} \tag{2.2.18.6}$$

$$= 20 - (-7) = 27 \text{ sq. units} \tag{2.2.18.7}$$

The following code generates Fig. 2.2.18.1

codes/parabola_line.py

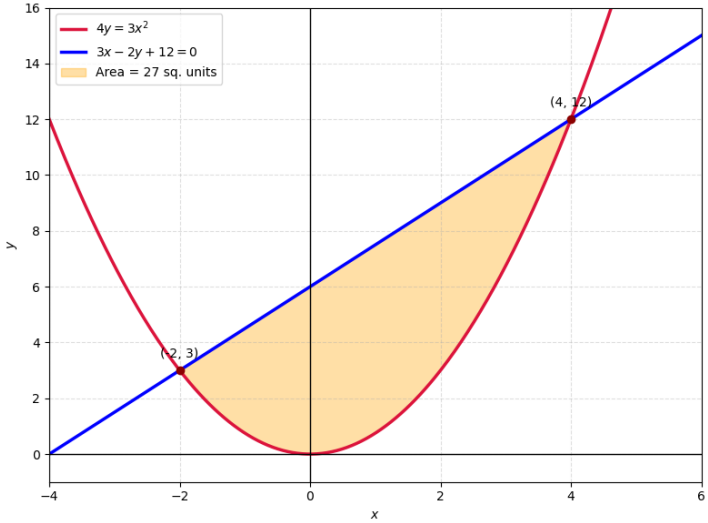


Fig. 2.2.18.1: Parabola and line

2.2.19 Consider $f : \mathbb{R} - \left\{-\frac{4}{3}\right\} \to \mathbb{R} - \left\{\frac{4}{3}\right\}$ given by $f(x) = \frac{4x+3}{3x+4}$. Show that f is bijective. Find the inverse of f and hence find $f^{-1}(0)$ and x such that $f^{-1}(x) = 2$. (12, 2017)

Solution: From (2.1.33.1), for one-one,

$$f(x_1) = f(x_2) \implies (4x_1 + 3)(3x_2 + 4) = (4x_2 + 3)(3x_1 + 4) \tag{2.2.19.1}$$

$$\implies 7x_1 = 7x_2 \implies x_1 = x_2 \tag{2.2.19.2}$$

For onto, let $y = \frac{4x+3}{3x+4}$ with $y \neq \frac{4}{3}$. Solving for x,

$$x = \frac{3 - 4y}{3y - 4} \tag{2.2.19.3}$$

which is defined for every $y \neq \frac{4}{3}$, and $x \neq -\frac{4}{3}$, since $x = -\frac{4}{3}$ would give $-9 = -16$.

So f is onto and hence bijective. The inverse is

$$f^{-1}(x) = \frac{3-4x}{3x-4} \tag{2.2.19.4}$$

From (2.1.34), $\mathbf{M} = \begin{pmatrix} 4 & 3 \\ 3 & 4 \end{pmatrix}$ with $|\mathbf{M}| = 7 \neq 0$, and $\mathbf{M}^{-1} = \frac{1}{7}\begin{pmatrix} 4 & -3 \\ -3 & 4 \end{pmatrix}$ gives the same inverse $\frac{4x-3}{-3x+4}$. Then

$$f^{-1}(0) = \frac{3}{-4} = -\frac{3}{4} \tag{2.2.19.5}$$

and

$$f^{-1}(x) = 2 \implies 3-4x = 6x-8 \implies x = \frac{11}{10} \tag{2.2.19.6}$$

2.2.20 Let $A = \mathbb{Q} \times \mathbb{Q}$ and let $*$ be a binary operation on A defined by $(a,b) * (c,d) = (ac, b+ad)$ for $(a,b), (c,d) \in A$. Determine, whether $*$ is commutative and associative. Then, with respect to $*$ on A

a) Find the identity element in A.
b) Find the invertible elements of A.

(12, 2017)

Solution: Writing (a,b) as the matrix $\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$, the product

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\begin{pmatrix} c & d \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} ac & ad+b \\ 0 & 1 \end{pmatrix} \tag{2.2.20.1}$$

is exactly $(a,b) * (c,d)$.

a) **Commutative:** $(a,b) * (c,d) = (ac, b+ad)$ whereas $(c,d) * (a,b) = (ac, d+cb)$. For $(1,2) * (3,4) = (3,6)$ and $(3,4) * (1,2) = (3,10)$. So $*$ is not commutative.
b) **Associative:**

$$[(a,b) * (c,d)] * (e,f) = (ac, b+ad) * (e,f) = (ace, b+ad+acf) \tag{2.2.20.2}$$
$$(a,b) * [(c,d) * (e,f)] = (a,b) * (ce, d+cf) = (ace, b+ad+acf) \tag{2.2.20.3}$$

The two are equal, so $*$ is associative.
c) **Identity:** Let (e_1, e_2) be the identity. Then $(a,b) * (e_1, e_2) = (ae_1, b+ae_2) = (a,b)$ for all a, b, giving $e_1 = 1$, $e_2 = 0$. The identity element is $(1,0)$, which also satisfies $(1,0) * (a,b) = (a,b)$.
d) **Invertible elements:** For (c,d) to be the inverse of (a,b),

$$(ac, b+ad) = (1,0) \implies c = \frac{1}{a}, \quad d = -\frac{b}{a} \tag{2.2.20.4}$$

which requires $a \neq 0$. So the invertible elements are (a,b) with $a \neq 0$, and the inverse is $\left(\frac{1}{a}, -\frac{b}{a}\right)$. In matrix form, this is $\left|\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}\right| = a \neq 0$.

2.2.21 If $\mathbf{A} = \begin{pmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{pmatrix}$, find $\mathbf{A}^{-1}$. Hence using $\mathbf{A}^{-1}$ solve the system of equations $2x - 3y + 5z = 11$, $3x + 2y - 4z = -5$, $x + y - 2z = -3$. (12, 2017)

Solution: Row reducing the augmented matrix $[\mathbf{A} \mid \mathbf{I}]$,

$$[\mathbf{A} \mid \mathbf{I}] = \begin{pmatrix} 2 & -3 & 5 & 1 & 0 & 0 \\ 3 & 2 & -4 & 0 & 1 & 0 \\ 1 & 1 & -2 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 3 & 2 & -4 & 0 & 1 & 0 \\ 2 & -3 & 5 & 1 & 0 & 0 \end{pmatrix} \tag{2.2.21.1}$$

$$\xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 - 3R_1} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 0 & -1 & 2 & 0 & 1 & -3 \\ 0 & -5 & 9 & 1 & 0 & -2 \end{pmatrix} \xrightarrow[R_2 \to -R_2]{R_3 \to R_3 - 5R_2'} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 0 & 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & -1 & 1 & -5 & 13 \end{pmatrix} \tag{2.2.21.2}$$

$$\xrightarrow{R_3 \to -R_3} \begin{pmatrix} 1 & 1 & -2 & 0 & 0 & 1 \\ 0 & 1 & -2 & 0 & -1 & 3 \\ 0 & 0 & 1 & -1 & 5 & -13 \end{pmatrix} \xrightarrow[R_1 \to R_1 + 2R_3]{R_2 \to R_2 + 2R_3} \begin{pmatrix} 1 & 1 & 0 & -2 & 10 & -25 \\ 0 & 1 & 0 & -2 & 9 & -23 \\ 0 & 0 & 1 & -1 & 5 & -13 \end{pmatrix} \tag{2.2.21.3}$$

$$\xrightarrow{R_1 \to R_1 - R_2} \begin{pmatrix} 1 & 0 & 0 & 0 & 1 & -2 \\ 0 & 1 & 0 & -2 & 9 & -23 \\ 0 & 0 & 1 & -1 & 5 & -13 \end{pmatrix} = [\mathbf{I} \mid \mathbf{A}^{-1}] \tag{2.2.21.4}$$

From (2.1.35.1),

$$\mathbf{A}^{-1} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix} \tag{2.2.21.5}$$

The system is $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 11 \\ -5 \\ -3 \end{pmatrix}$. Since $\text{rank}\,(\mathbf{A}) = \text{rank}\,([\mathbf{A} \mid \mathbf{b}]) = 3$, from (2.1.26.2) the solution is unique. From (2.1.35.1),

$$\mathbf{x} = \mathbf{A}^{-1}\mathbf{b} = \begin{pmatrix} 0 & 1 & -2 \\ -2 & 9 & -23 \\ -1 & 5 & -13 \end{pmatrix} \begin{pmatrix} 11 \\ -5 \\ -3 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \tag{2.2.21.6}$$

Therefore,

$$x = 1, \quad y = 2, \quad z = 3 \tag{2.2.21.7}$$

The following code verifies the solution

codes/inverse.py